



US012389346B2

(12) **United States Patent**
Cui et al.

(10) **Patent No.:** **US 12,389,346 B2**

(45) **Date of Patent:** **Aug. 12, 2025**

(54) **ENHANCED RADIO RESOURCE
MANAGEMENT FOR SYNCHRONIZATION
SIGNAL BLOCKS OUTSIDE AN ACTIVE
BANDWIDTH PART**

(58) **Field of Classification Search**

None

See application file for complete search history.

(71) Applicant: **Apple Inc.**, Cupertino, CA (US)

(56) **References Cited**

U.S. PATENT DOCUMENTS

2023/0117005 A1 * 4/2023 Ang H04L 5/0005
370/331

2023/0144052 A1 * 5/2023 Islam H04W 74/0833
370/329

FOREIGN PATENT DOCUMENTS

WO WO-2022021031 A1 * 2/2022 H04L 5/0053

OTHER PUBLICATIONS

[No Author Listed], "3rd Generation Partnership Project; Technical Specification Group Radio Access Network; NR; Requirements for support of radio resource management (Release 17)," 3GPP TS 38.133 V17.3.0, Sep. 2021, 3209 pages.

(Continued)

Primary Examiner — Ayanah S George

(74) *Attorney, Agent, or Firm* — Fish & Richardson P.C.

(73) Assignee: **Apple Inc.**, Cupertino, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 723 days.

(21) Appl. No.: **17/553,492**

(22) Filed: **Dec. 16, 2021**

(65) **Prior Publication Data**

US 2023/0199678 A1 Jun. 22, 2023

(51) **Int. Cl.**

H04W 56/00 (2009.01)

H04W 24/10 (2009.01)

H04W 72/044 (2023.01)

H04W 72/23 (2023.01)

H04W 72/51 (2023.01)

(52) **U.S. Cl.**

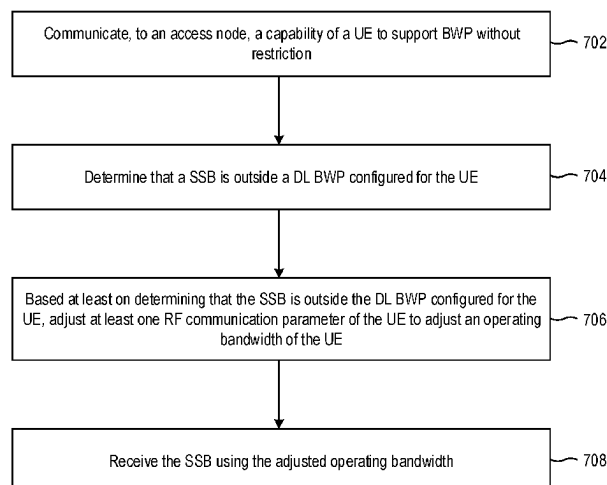
CPC **H04W 56/001** (2013.01); **H04W 24/10**
(2013.01); **H04W 72/044** (2013.01); **H04W**
72/23 (2023.01); **H04W 72/51** (2023.01)

(57) **ABSTRACT**

A method performed by a user equipment (UE) for enhanced radio resource management (RRM) includes communicating, to an access node, a capability of the UE to support bandwidth part (BWP) without restriction. The UE determines that a synchronization signal block (SSB) is outside a downlink (DL) BWP configured for the UE. Based on determining that the SSB is outside the DL BWP configured for the UE, at least one radio frequency (RF) communication parameter of the UE is adjusted to adjust an operating bandwidth of the UE, and the SSB is received using the adjusted operating bandwidth.

18 Claims, 10 Drawing Sheets

700 ↘



(56)

References Cited

OTHER PUBLICATIONS

[No Author Listed], "3rd Generation Partnership Project; Technical Specification Group Radio Access Network; NR; Radio Resource Control (RRC) protocol specification (Release 16)," 3GPP TS 38.331 V16.6.0, Sep. 2021, 961 pages.

[No Author Listed], "3rd Generation Partnership Project; Technical Specification Group Radio Access Network; NR; User Equipment (UE) radio access capabilities (Release 16)," 3GPP TS 38.306 V16.6.0, Sep. 2021, 155 pages.

* cited by examiner

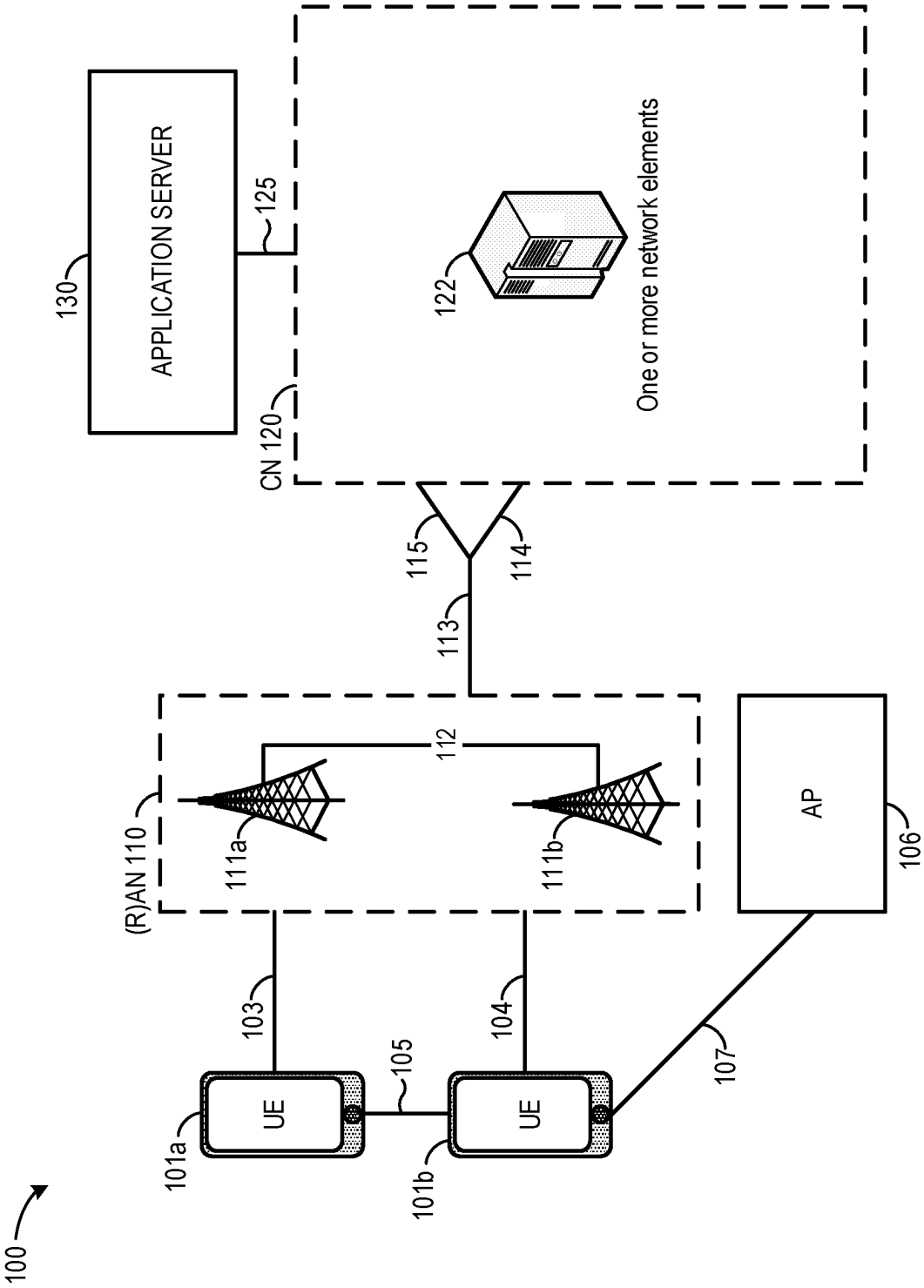


FIG. 1

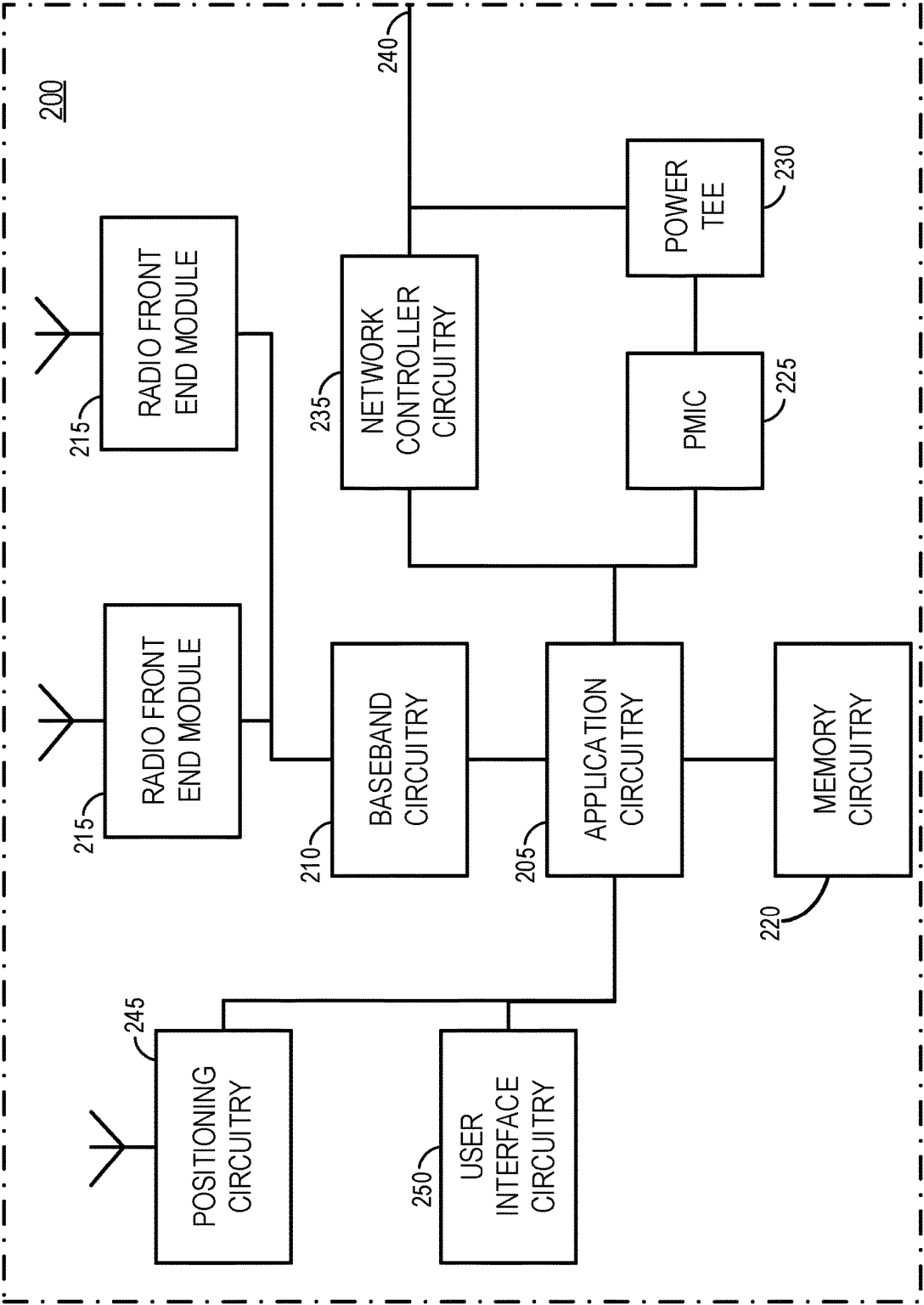


FIG. 2

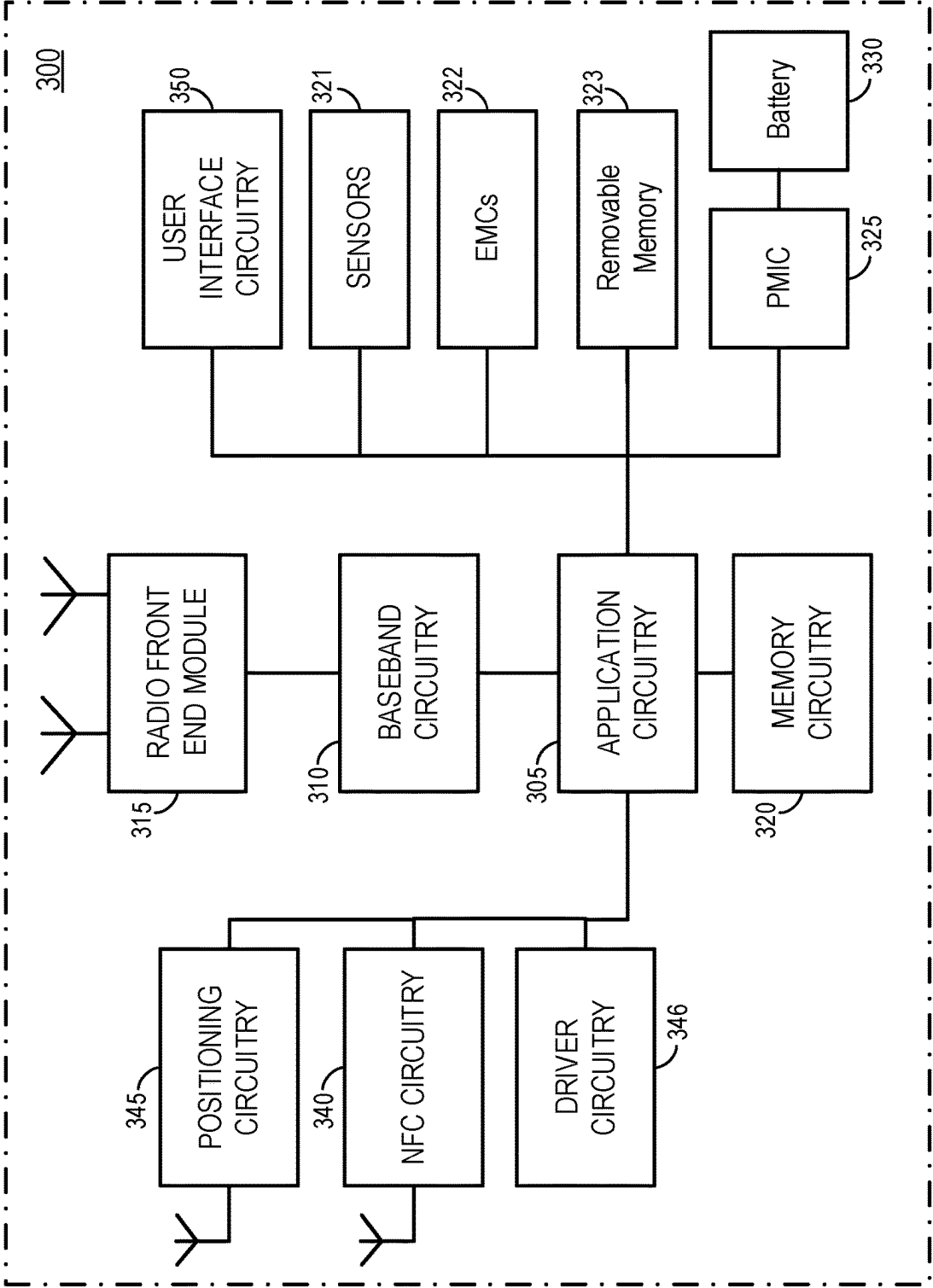


FIG. 3

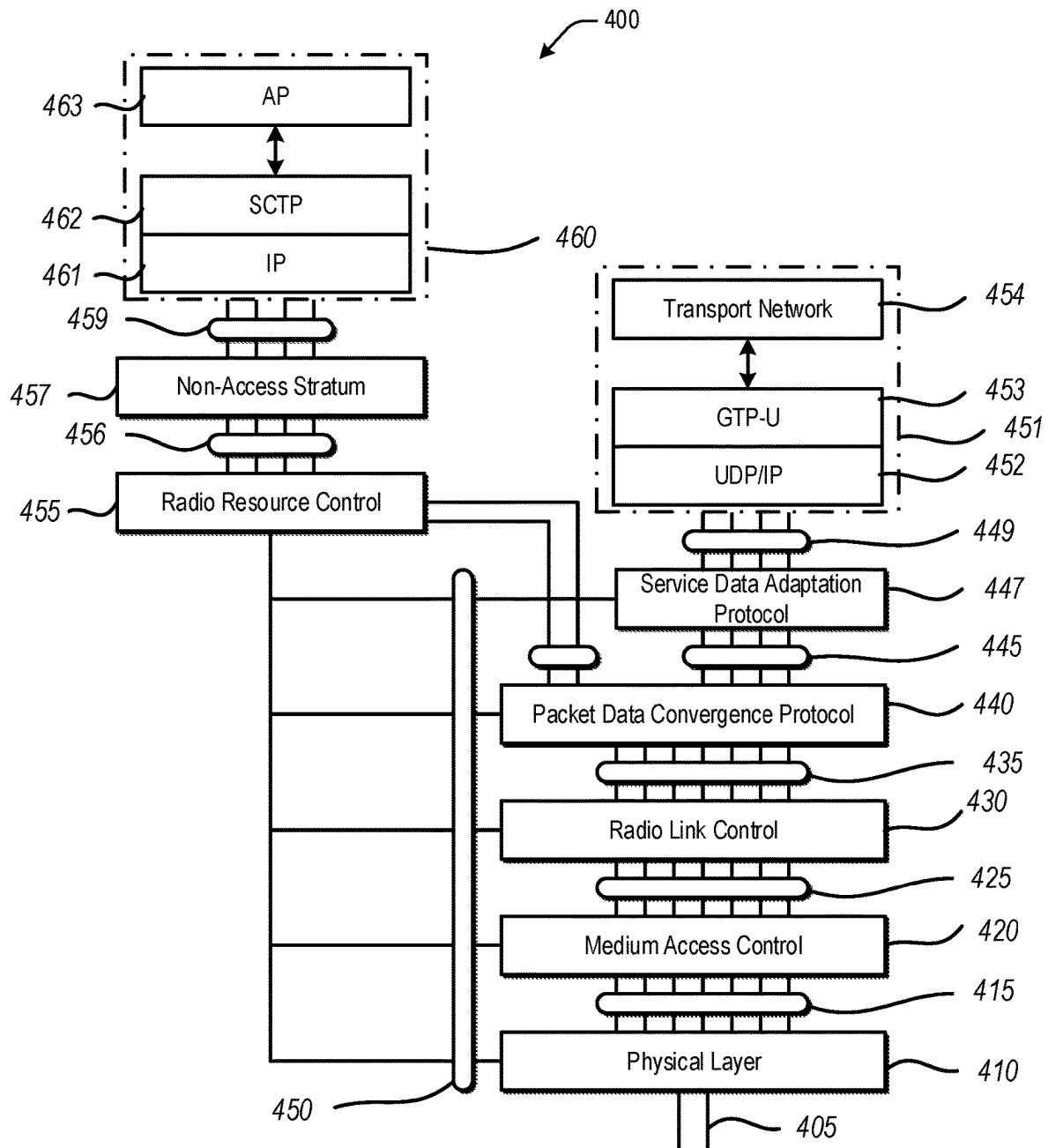


FIG. 4

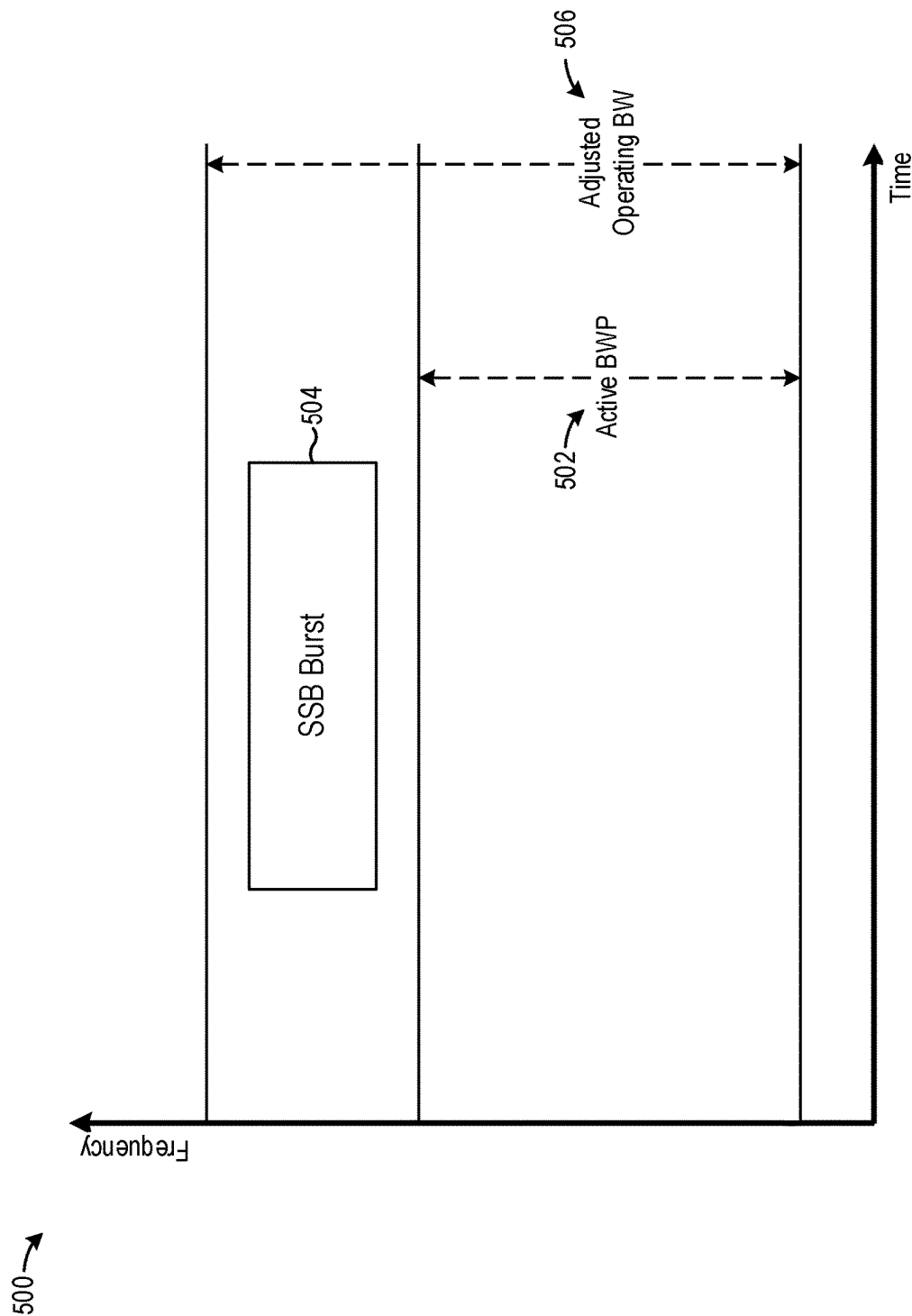


FIG. 5A

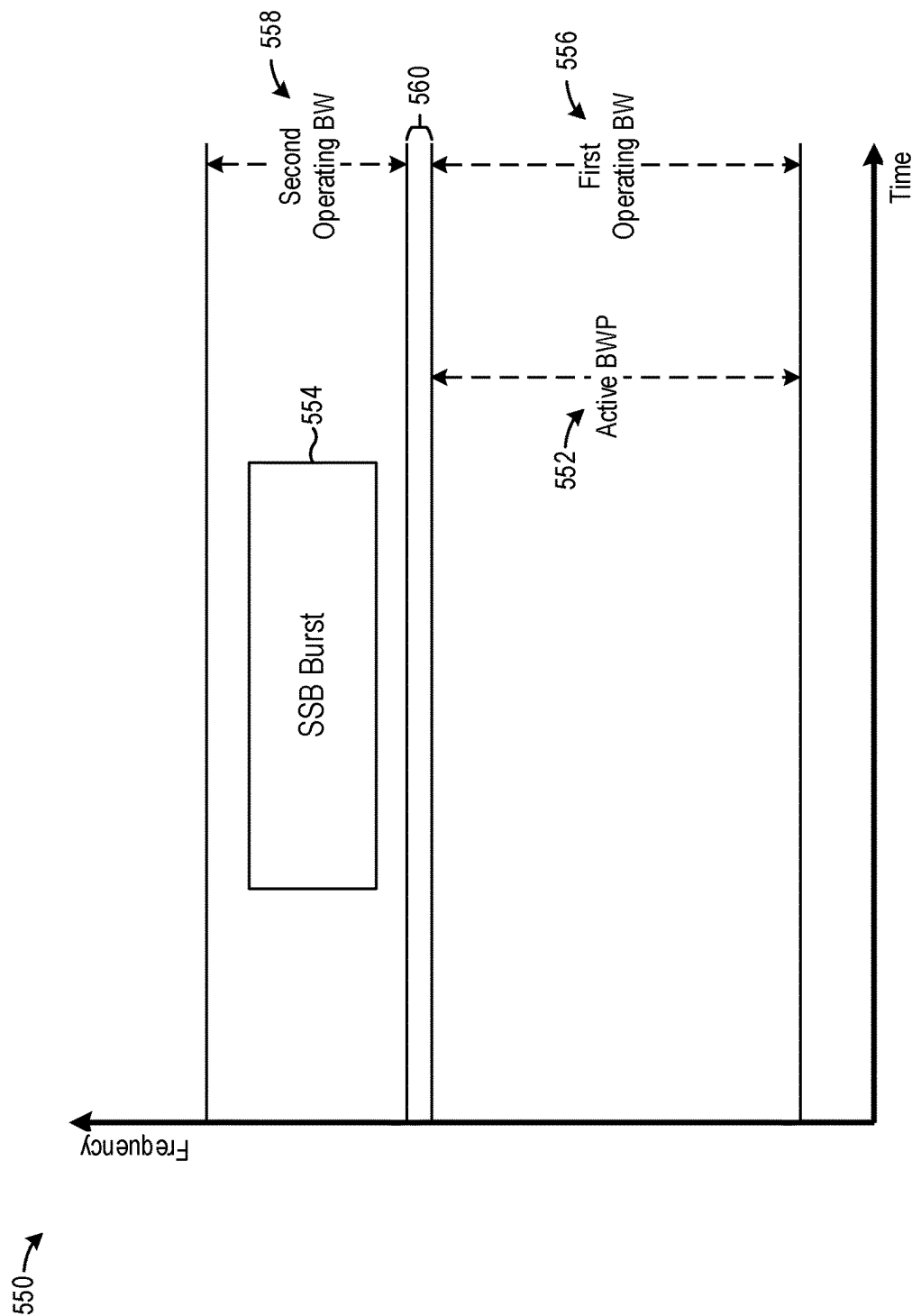


FIG. 5B

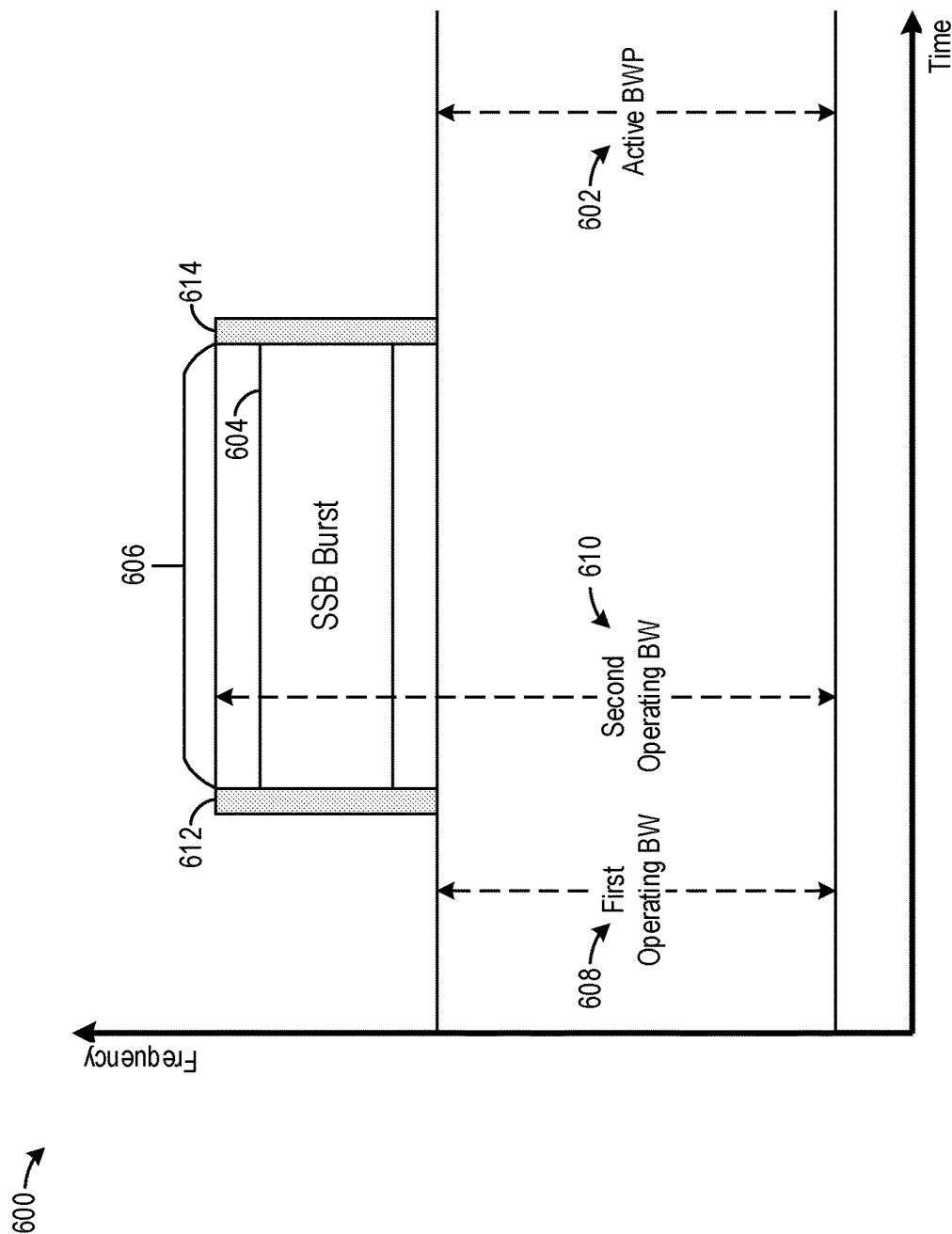


FIG. 6A

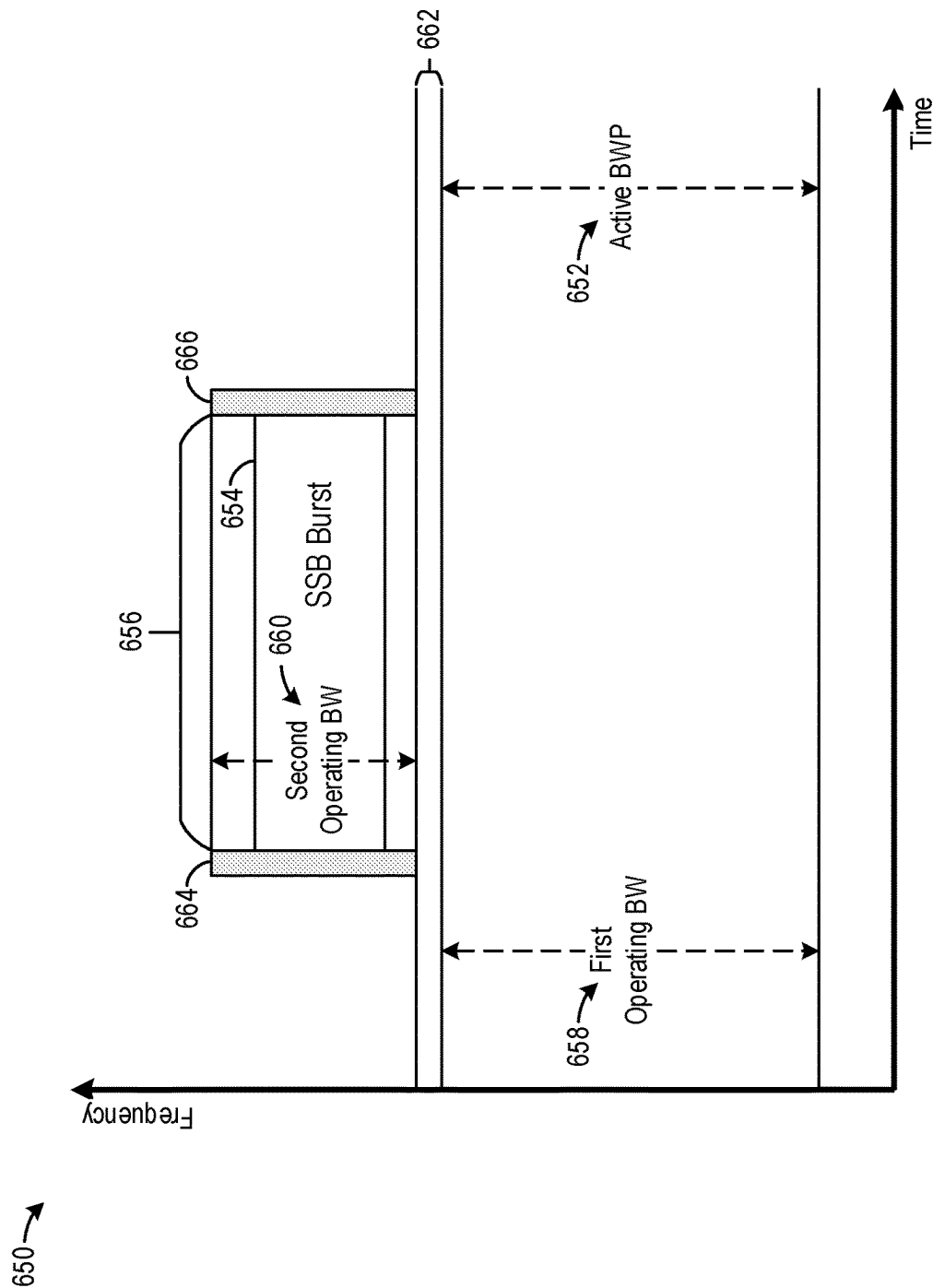
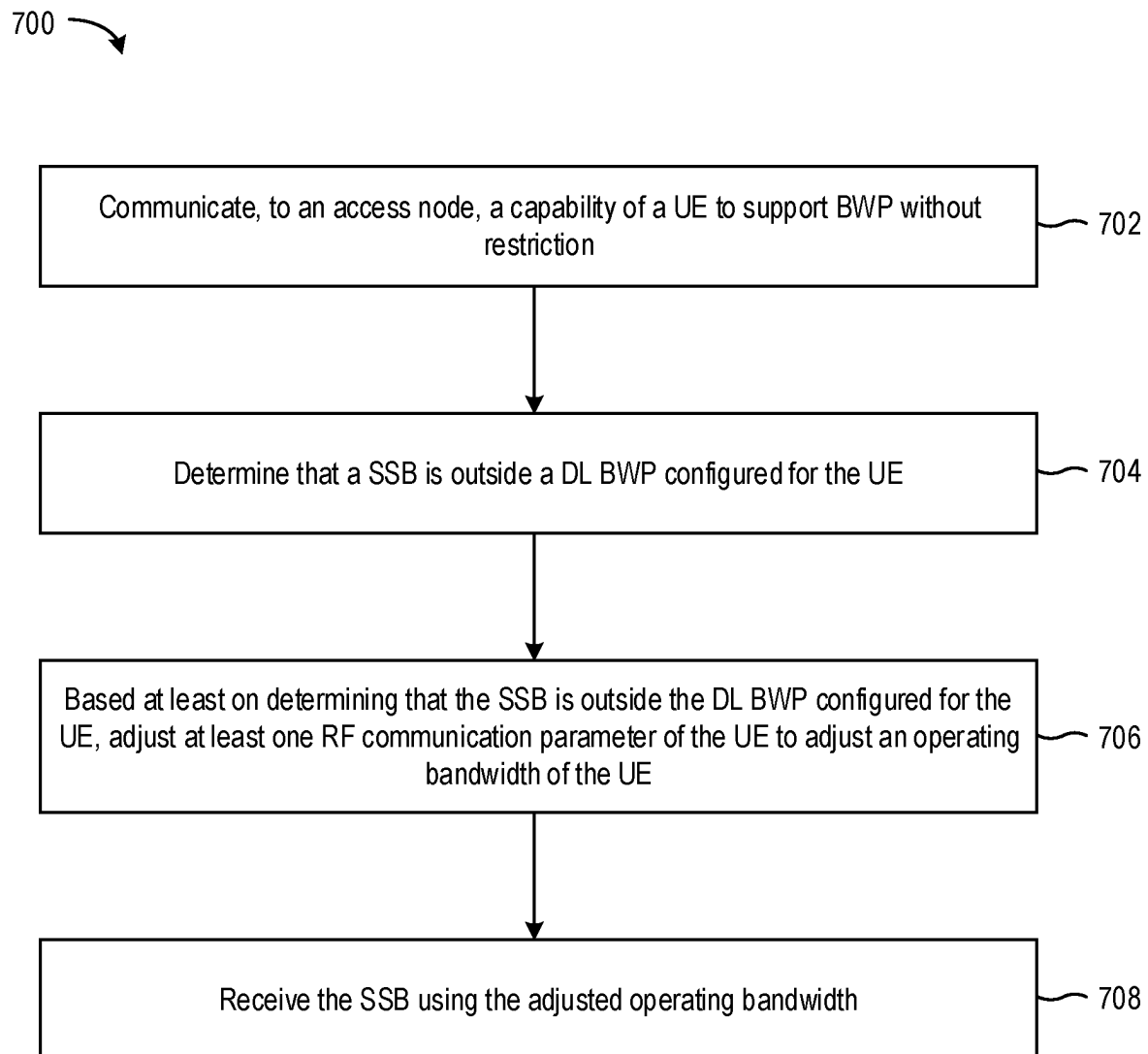


FIG. 6B

**FIG. 7**

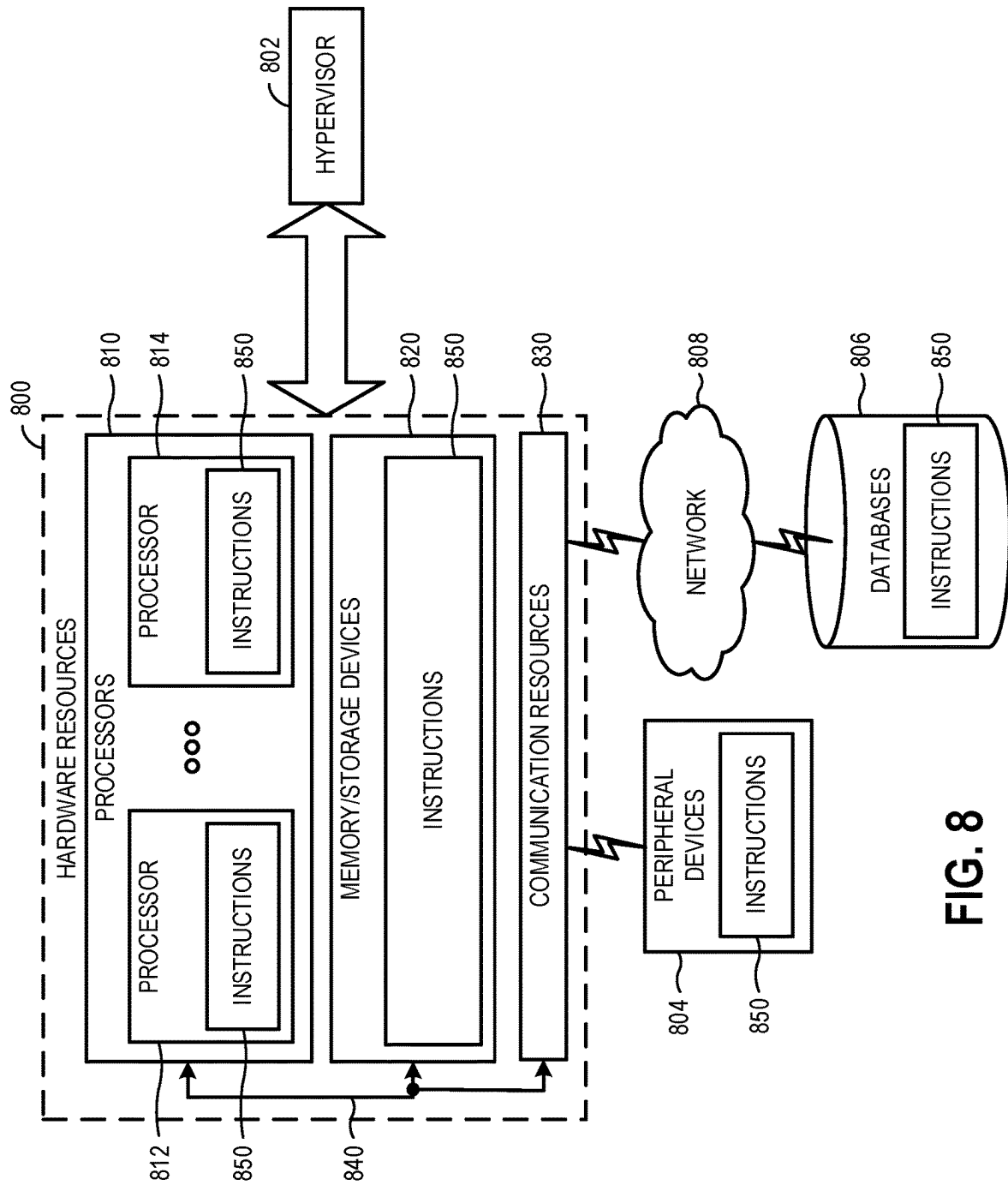


FIG. 8

1

ENHANCED RADIO RESOURCE MANAGEMENT FOR SYNCHRONIZATION SIGNAL BLOCKS OUTSIDE AN ACTIVE BANDWIDTH PART

TECHNICAL FIELD

This disclosure relates to techniques for enhancing radio resource management, particularly for synchronization signal blocks outside an active bandwidth part.

BACKGROUND

In wireless communication networks, two or more devices communicate with each other over a communication channel having a bandwidth that spans a range of frequencies. One way to increase the capacity and throughput of these networks is to increase the bandwidth of the channel. For example, to increase the capacity and throughput of cellular networks (among other things), the Third Generation Partnership Project (3GPP) increased the channel bandwidth from 20 MHz in Long-Term Evolution (LTE) networks to 100 MHz (for frequency range 1) or 400 MHz (for frequency range 2) in Fifth Generation (5G) New Radio (NR) networks.

However, some devices may not be capable of operating on the full bandwidth used by, for example, a 5G NR network. Even if a device is capable of utilizing the entire bandwidth, it may not be efficient for the device to do so, as monitoring the entire bandwidth can increase device power consumption. To address these and other concerns, the 5G NR standard allows for a device to be configured to operate on a subset of the total channel bandwidth, referred to as a bandwidth part (BWP). Once configured, the device can transmit and receive data, carry out radio resource management (RRM) activities, and perform other operations within the active BWP.

SUMMARY

The technology described here enables performance of RRM activities outside of an active downlink (DL) BWP when a user equipment (UE) indicates support for BWP operation without bandwidth restriction. In particular, techniques for configuring a UE to receive synchronization signal blocks (SSBs) outside of an active DL BWP are described. In addition, ways in which a network (e.g., an access node) can interpret an indication of support for BWP operation without bandwidth restriction, including whether the UE requires a measurement gap or interruption(s) for RF adjustment, are provided. In this manner, the network and the UE can coordinate RRM activities including intra-frequency and serving cell measurements, SCell activation procedures, and PSCell addition, among others, outside of an active DL BWP, thereby improving RRM flexibility and performance.

In general, in a first aspect, a method performed by a UE, includes: communicating, to an access node, a capability of the UE to support BWP without restriction, determining that a SSB is outside a DL BWP configured for the UE, adjusting at least one RF communication parameter of the UE to adjust an operating bandwidth of the UE based at least on determining that the SSB is outside the DL BWP configured for the UE, and receiving the SSB using the adjusted operating bandwidth. In some examples, a system or device is configured to perform the method. In some examples, one or

2

more non-transitory computer-readable storage mediums store instructions executable by at least one processor to perform the method.

In a second aspect combinable with the first aspect, adjusting the at least one RF communication parameter of the UE to adjust the operating bandwidth of the UE includes configuring the operating bandwidth to include the DL BWP and an additional bandwidth that includes the SSB.

In a third aspect combinable with the first or second aspects, configuring the operating bandwidth to include the DL BWP and the additional bandwidth that includes the SSB includes determining the additional bandwidth that includes the SSB, and expanding the operating bandwidth from a first bandwidth that includes the DL BWP to a second bandwidth that includes the DL BWP and the additional bandwidth.

In a fourth aspect combinable with any of the first through third aspects, adjusting the at least one RF communication parameter of the UE to adjust the operating bandwidth of the UE includes configuring a second operating bandwidth for the UE that is in addition to a first operating bandwidth of the UE, in which the first operating bandwidth includes the DL BWP and the second operating bandwidth includes the SSB.

In a fifth aspect combinable with any of the first through fourth aspects, configuring the second operating bandwidth includes determining an additional bandwidth that includes the SSB, and configuring the second operating bandwidth to include the additional bandwidth.

In a sixth aspect combinable with any of the first through fifth aspects, communicating to the access node the capability of the UE to support BWP without restriction includes sending a bwp-WithoutRestriction information element (IE) to the access node.

In a seventh aspect combinable with any of the first through sixth aspects, adjusting the at least one RF communication parameter of the UE to adjust the operating bandwidth of the UE includes adjusting the at least one RF communication parameter upon occurrence of a SSB measurement window for the SSB.

In an eighth aspect combinable with any of the first through seventh aspects, the operating bandwidth of the UE prior to the occurrence of the SSB measurement window corresponds to a first bandwidth, the method further including: upon expiration of the SSB measurement window, readjusting the at least one RF communication parameter to revert the operating bandwidth of the UE to the first bandwidth.

In a ninth aspect combinable with any of the first through eighth aspects, the method further include communicating, to the access node, an indication that a measurement gap is needed by the UE for SSB measurement, in which adjusting the at least one RF communication parameter of the UE to adjust the operating bandwidth of the UE includes adjusting the at least one RF communication parameter upon occurrence of the measurement gap.

In a tenth aspect combinable with any of the first through ninth aspects, communicating to the access node the indication that the measurement gap is needed by the UE includes communicating to the access node a requirement for the measurement gap using a NeedForGapsIntraFreq-r16 IE.

In an eleventh aspect combinable with any of the first through tenth aspects, the SSB received using the adjusted operating bandwidth is associated with a target secondary cell (SCell), and the adjusted operating bandwidth is not readjusted during a SCell activation procedure period for the target SCell.

3

In a twelfth aspect combinable with any of the first through eleventh aspects, the SSB received using the adjusted operating bandwidth is associated with a target SCell, and the operating bandwidth of the UE prior to adjustment corresponds to a first bandwidth, the method further including readjusting the at least one RF communication parameter to revert the operating bandwidth of the UE to the first bandwidth during a SCell activation procedure period for the target SCell.

In a thirteenth aspect combinable with any of the first through twelfth aspects, determining that the SSB is outside the DL BWP configured for the UE is based on an indication from the access node, the indication including a center frequency for the SSB.

In general, in a fourteenth aspect, a method performed by an access node includes receiving, from a device, an indication of a capability of the device to support BWP without restriction, and in response to the indication, configuring the device to receive a SSB that is outside a DL BWP configured for the device, the configuring including causing the device to adjust at least one RF communication parameter of the device to adjust an operating bandwidth of the device. In some examples, a system or device is configured to perform the method. In some examples, one or more non-transitory computer-readable storage mediums store instructions executable by at least one processor to perform the method.

In general, in a fifteenth aspect combinable with the fourteenth aspect, receiving the indication of the capability of the device to support BWP without restriction includes receiving a bwp-WithoutRestriction IE from the device.

In general, in a sixteenth aspect combinable with the fourteenth or fifteenth aspects, the method includes causing an interruption period before and after a measurement occasion for the SSB that is outside the DL BWP configured for the device based at least in part on the indication of the capability of the device to support BWP without restriction.

In general, in a seventeenth aspect combinable with any of the fourteenth through sixteenth aspects, the method includes receiving, from the device, an indication that a measurement gap is needed by the device for receipt of the SSB that is outside the DL BWP configured for the device, and scheduling the measurement gap for the device before and after a measurement occasion for the SSB that is outside the DL BWP configured for the device.

The details of one or more implementations are set forth in the accompanying drawings and the description below. Other features and advantages will be apparent from the description and drawings, and from the claims.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 illustrates an example of a wireless communication system.

FIG. 2 illustrates an example of infrastructure equipment.

FIG. 3 illustrates an example of a device.

FIG. 4 illustrates various protocol functions.

FIGS. 5A and 5B illustrate examples of BWP operation without bandwidth restriction.

FIGS. 6A and 6B illustrate examples BWP operation without bandwidth restriction.

FIG. 7 illustrates an example process for enhanced RRM for SSBs outside an active BWP.

FIG. 8 illustrates an example computer system.

Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

FIG. 1 illustrates an example of a wireless communication system **100**. For purposes of convenience and without limi-

4

tation, the example system **100** is described in the context of Long Term Evolution (LTE) and Fifth Generation (5G) New Radio (NR) communication standards as defined by the Third Generation Partnership Project (3GPP) technical specifications. More specifically, the wireless communication system **100** is described in the context of a Non-Standalone (NSA) networks that incorporate both LTE and NR, for example, E-UTRA (Evolved Universal Terrestrial Radio Access)-NR Dual Connectivity (EN-DC) networks, and NE-DC networks. However, the wireless communication system **100** may also be a Standalone (SA) network that incorporates only NR. Furthermore, other types of communication standards are possible, including future 3GPP systems (e.g., Sixth Generation (6G)) systems, IEEE 802.16 protocols (e.g., WMAN, WiMAX, etc.), or the like.

As shown by FIG. 1, the system **100** includes UE **101a** and UE **101b** (collectively referred to as “UEs **101**” or “UE **101**”). In this example, UEs **101** are illustrated as smart-phones (e.g., handheld touchscreen mobile computing devices connectable to one or more cellular networks), but may also comprise any mobile or non-mobile computing device, such as consumer electronics devices, cellular phones, smartphones, feature phones, tablet computers, wearable computer devices, personal digital assistants (PDAs), pagers, wireless handsets, desktop computers, laptop computers, in-vehicle infotainment (IVI), in-car entertainment (ICE) devices, an Instrument Cluster (IC), head-up display (HUD) devices, onboard diagnostic (OBD) devices, dashtop mobile equipment (DME), mobile data terminals (MDTs), Electronic Engine Management System (EEMS), electronic/engine control units (ECUs), electronic/engine control modules (ECMs), embedded systems, microcontrollers, control modules, engine management systems (EMS), networked or “smart” appliances, MTC devices, M2M, IoT devices, and/or the like.

In some examples, any of the UEs **101** may be IoT UEs, which may comprise a network access layer designed for low-power IoT applications utilizing short-lived UE connections. An IoT UE can utilize technologies such as M2M or MTC for exchanging data with an MTC server or device via a PLMN, ProSe or D2D communication, sensor networks, or IoT networks. The M2M or MTC exchange of data may be a machine-initiated exchange of data. An IoT network describes interconnecting IoT UEs, which may include uniquely identifiable embedded computing devices (within the Internet infrastructure), with short-lived connections. The IoT UEs may execute background applications (e.g., keep-alive messages, status updates, etc.) to facilitate the connections of the IoT network.

The UEs **101** may be configured to connect, for example, communicatively couple, with RAN **110**. In some examples, the RAN **110** may be an NG RAN or a 5G RAN, an E-UTRAN, or a legacy RAN, such as a UTRAN or GERAN. As used herein, the term “NG RAN” or the like may refer to a RAN **110** that operates in an NR or 5G system **100**, and the term “E-UTRAN” or the like may refer to a RAN **110** that operates in an LTE or 4G system **100**. The UEs **101** utilize connections (or channels) **103** and **104**, respectively, each of which comprises a physical communications interface or layer (discussed in further detail below).

In this example, the connections **103** and **104** are illustrated as an air interface to enable communicative coupling, and can be consistent with cellular communications protocols, such as a GSM protocol, a CDMA network protocol, a PTT protocol, a POC protocol, a UMTS protocol, a 3GPP LTE protocol, an Advanced long term evolution (LTE-A) protocol, a LTE-based access to unlicensed spectrum (LTE-

U), a 5G protocol, a NR protocol, an NR-based access to unlicensed spectrum (NR-U) protocol, and/or any of the other communications protocols discussed herein. In some examples, the UEs 101 may directly exchange communication data via a ProSe interface 105. The ProSe interface 105 may alternatively be referred to as a SL interface 105 and may comprise one or more logical channels, including but not limited to a PSCCH, a PSSCH, a PSDCH, and a PSBCH.

The UE 101b is shown to be configured to access an AP 106 (also referred to as “WLAN node 106,” “WLAN 106,” “WLAN Termination 106,” “WT 106” or the like) via connection 107. The connection 107 can comprise a local wireless connection, such as a connection consistent with any IEEE 802.11 protocol, wherein the AP 106 would comprise a wireless fidelity (Wi-Fi®) router. In this example, the AP 106 is shown to be connected to the Internet without connecting to the core network of the wireless system (described in further detail below). In various examples, the UE 101b, RAN 110, and AP 106 may be configured to utilize LWA operation and/or LWIP operation. The LWA operation may involve the UE 101b in RRC CONNECTED being configured by a RAN node 111a-b to utilize radio resources of LTE and WLAN. LWIP operation may involve the UE 101b using WLAN radio resources (e.g., connection 107) via IPsec protocol tunneling to authenticate and encrypt packets (e.g., IP packets) sent over the connection 107. IPsec tunneling may include encapsulating the entirety of original IP packets and adding a new packet header, thereby protecting the original header of the IP packets.

The RAN 110 can include one or more AN nodes or RAN nodes 111a and 111b (collectively referred to as “RAN nodes 111” or “RAN node 111”) that enable the connections 103 and 104. As used herein, the terms “access node,” “access point,” or the like may describe equipment that provides the radio baseband functions for data and/or voice connectivity between a network and one or more users. These access nodes can be referred to as BS, gNBs, RAN nodes, eNBs, NodeBs, RSUs, TRxPs or TRPs, and so forth, and can comprise ground stations (e.g., terrestrial access points) or satellite stations providing coverage within a geographic area (e.g., a cell). As used herein, the term “NG RAN node” or the like may refer to a RAN node 111 that operates in an NR or 5G system 100 (for example, a gNB), and the term “E-UTRAN node” or the like may refer to a RAN node 111 that operates in an LTE or 4G system 100 (e.g., an eNB). According to various examples, the RAN nodes 111 may be implemented as one or more of a dedicated physical device such as a macrocell base station, and/or a low power (LP) base station for providing femto-cells, picocells or other like cells having smaller coverage areas, smaller user capacity, or higher bandwidth compared to macrocells.

In some examples, all or parts of the RAN nodes 111 may be implemented as one or more software entities running on server computers as part of a virtual network, which may be referred to as a CRAN and/or a virtual baseband unit pool (vBBUP). In these examples, the CRAN or vBBUP may implement a RAN function split, such as a PDCP split wherein RRC and PDCP layers are operated by the CRAN/vBBUP and other L2 protocol entities are operated by individual RAN nodes 111; a MAC/PHY split wherein RRC, PDCP, RLC, and MAC layers are operated by the CRAN/vBBUP and the PHY layer is operated by individual RAN nodes 111; or a “lower PHY” split wherein RRC, PDCP, RLC, MAC layers and upper portions of the PHY layer are operated by the CRAN/vBBUP and lower portions of the

PHY layer are operated by individual RAN nodes 111. This virtualized framework allows the freed-up processor cores of the RAN nodes 111 to perform other virtualized applications. In some implementations, an individual RAN node 111 may represent individual gNB-DUs that are connected to a gNB-CU via individual F1 interfaces (not shown by FIG. 1). In these implementations, the gNB-DUs may include one or more remote radio heads or RFEMs (see, e.g., FIG. 2), and the gNB-CU may be operated by a server that is located in the RAN 110 (not shown) or by a server pool in a similar manner as the CRAN/vBBUP. Additionally or alternatively, one or more of the RAN nodes 111 may be next generation eNBs (ng-eNBs), which are RAN nodes that provide E-UTRA user plane and control plane protocol terminations toward the UEs 101, and are connected to a 5GC via an NG interface (discussed infra).

Any of the RAN nodes 111 can terminate the air interface protocol and can be the first point of contact for the UEs 101. In some examples, any of the RAN nodes 111 can fulfill various logical functions for the RAN 110 including, but not limited to, radio network controller (RNC) functions such as radio bearer management, uplink and downlink dynamic radio resource management and data packet scheduling, and mobility management.

In some examples, the UEs 101 can be configured to communicate using OFDM communication signals with each other or with any of the RAN nodes 111 over a multicarrier communication channel in accordance with various communication techniques, such as, but not limited to, an OFDMA communication technique (e.g., for downlink communications) or a SC-FDMA communication technique (e.g., for uplink and ProSe or sidelink communications), although the scope of the examples is not limited in this respect. The OFDM signals can comprise a plurality of orthogonal subcarriers.

In some examples, a downlink resource grid can be used for downlink transmissions from any of the RAN nodes 111 to the UEs 101, while uplink transmissions can utilize similar techniques. The grid can be a time-frequency grid, called a resource grid or time-frequency resource grid, which is the physical resource in the downlink in each slot. Such a time-frequency plane representation is a common practice for OFDM systems, which makes it intuitive for radio resource allocation. Each column and each row of the resource grid corresponds to one OFDM symbol and one OFDM subcarrier, respectively. The duration of the resource grid in the time domain corresponds to one slot in a radio frame. The smallest time-frequency unit in a resource grid is denoted as a resource element. Each resource grid comprises a number of resource blocks, which describe the mapping of certain physical channels to resource elements. Each resource block comprises a collection of resource elements; in the frequency domain, this may represent the smallest quantity of resources that currently can be allocated. There are several different physical downlink channels that are conveyed using such resource blocks.

According to various examples, the UEs 101 and the RAN nodes 111 communicate data (for example, transmit and receive) data over a licensed medium (also referred to as the “licensed spectrum” and/or the “licensed band”) and an unlicensed shared medium (also referred to as the “unlicensed spectrum” and/or the “unlicensed band”). The licensed spectrum may include channels that operate in the frequency range of approximately 400 MHz to approximately 3.8 GHz, whereas the unlicensed spectrum may include the 5 GHz band. NR in the unlicensed spectrum may

be referred to as NR-U, and LTE in an unlicensed spectrum may be referred to as LTE-U, licensed assisted access (LAA), or MulteFire.

To operate in the unlicensed spectrum, the UEs **101** and the RAN nodes **111** may operate using LAA, eLAA, and/or feLAA mechanisms. In these implementations, the UEs **101** and the RAN nodes **111** may perform one or more known medium-sensing operations and/or carrier-sensing operations in order to determine whether one or more channels in the unlicensed spectrum is unavailable or otherwise occupied prior to transmitting in the unlicensed spectrum. The medium/carrier sensing operations may be performed according to a listen-before-talk (LBT) protocol.

LBT is a mechanism whereby equipment (for example, UEs **101** RAN nodes **111**, etc.) senses a medium (for example, a channel or carrier frequency) and transmits when the medium is sensed to be idle (or when a specific channel in the medium is sensed to be unoccupied). The medium sensing operation may include CCA, which utilizes at least ED to determine the presence or absence of other signals on a channel in order to determine if a channel is occupied or clear. This LBT mechanism allows cellular/LAA networks to coexist with incumbent systems in the unlicensed spectrum and with other LAA networks. ED may include sensing RF energy across an intended transmission band for a period of time and comparing the sensed RF energy to a predefined or configured threshold.

Typically, the incumbent systems in the 5 GHz band are WLANs based on IEEE 802.11 technologies. WLAN employs a contention-based channel access mechanism, called CSMA/CA. Here, when a WLAN node (e.g., a mobile station (MS) such as UE **101**, AP **106**, or the like) intends to transmit, the WLAN node may first perform CCA before transmission. Additionally, a backoff mechanism is used to avoid collisions in situations where more than one WLAN node senses the channel as idle and transmits at the same time. The backoff mechanism may be a counter that is drawn randomly within the CWS, which is increased exponentially upon the occurrence of collision and reset to a minimum value when the transmission succeeds. The LBT mechanism designed for LAA is somewhat similar to the CSMA/CA of WLAN. In some implementations, the LBT procedure for DL or UL transmission bursts including PDSCH or PUSCH transmissions, respectively, may have an LAA contention window that is variable in length between X and Y ECCA slots, where X and Y are minimum and maximum values for the CWSs for LAA. In one example, the minimum CWS for an LAA transmission may be 9 microseconds (s); however, the size of the CWS and a MCOT (for example, a transmission burst) may be based on governmental regulatory requirements.

The LAA mechanisms are built upon CA technologies of LTE-Advanced systems. In CA, each aggregated carrier is referred to as a CC. A CC may have a bandwidth of 1.4, 3, 5, 10, 15 or 20 MHz and a maximum of five CCs can be aggregated, and therefore, a maximum aggregated bandwidth is 100 MHz. In FDD systems, the number of aggregated carriers can be different for DL and UL, where the number of UL CCs is equal to or lower than the number of DL component carriers. In some cases, individual CCs can have a different bandwidth than other CCs. In TDD systems, the number of CCs as well as the bandwidths of each CC is usually the same for DL and UL.

CA also comprises individual serving cells to provide individual CCs. The coverage of the serving cells may differ, for example, because CCs on different frequency bands will experience different pathloss. A primary service cell (PCell)

may provide a PCC for both UL and DL, and may handle RRC and NAS related activities. The other serving cells are referred to as SCells, and each SCell may provide an individual SCC for both UL and DL. The SCCs may be added and removed as required, while changing the PCC may require the UE **101** to undergo a handover. In LAA, eLAA, and feLAA, some or all of the SCells may operate in the unlicensed spectrum (referred to as "LAA SCells"), and the LAA SCells are assisted by a PCell operating in the licensed spectrum. When a UE is configured with more than one LAA SCell, the UE may receive UL grants on the configured LAA SCells indicating different PUSCH starting positions within a same subframe.

The PDSCH carries user data and higher-layer signaling to the UEs **101**. The PDCCH carries information about the transport format and resource allocations related to the PDSCH channel, among other things. It may also inform the UEs **101** about the transport format, resource allocation, and HARQ information related to the uplink shared channel. Typically, downlink scheduling (assigning control and shared channel resource blocks to the UE **101b** within a cell) may be performed at any of the RAN nodes **111** based on channel quality information fed back from any of the UEs **101**. The downlink resource assignment information may be sent on the PDCCH used for (e.g., assigned to) each of the UEs **101**.

The PDCCH uses CCEs to convey the control information. Before being mapped to resource elements, the PDCCH complex-valued symbols may first be organized into quadruplets, which may then be permuted using a sub-block interleaver for rate matching. Each PDCCH may be transmitted using one or more of these CCEs, where each CCE may correspond to nine sets of four physical resource elements known as REGs. Four Quadrature Phase Shift Keying (QPSK) symbols may be mapped to each REG. The PDCCH can be transmitted using one or more CCEs, depending on the size of the DCI and the channel condition. There can be four or more different PDCCH formats defined in LTE with different numbers of CCEs (e.g., aggregation level, L=1, 2, 4, or 8).

Some examples may use concepts for resource allocation for control channel information that are an extension of the above-described concepts. For example, some examples may utilize an EPDCCH that uses PDSCH resources for control information transmission. The EPDCCH may be transmitted using one or more ECCEs. Similar to above, each ECCE may correspond to nine sets of four physical resource elements known as an EREGs. An ECCE may have other numbers of EREGs in some situations.

The RAN nodes **111** may be configured to communicate with one another via interface **112**. In examples where the system **100** is an LTE system, the interface **112** may be an X2 interface **112**. The X2 interface may be defined between two or more RAN nodes **111** (e.g., two or more eNBs and the like) that connect to EPC **120**, and/or between two eNBs connecting to EPC **120**. In some implementations, the X2 interface may include an X2 user plane interface (X2-U) and an X2 control plane interface (X2-C). The X2-U may provide flow control mechanisms for user data packets transferred over the X2 interface, and may be used to communicate information about the delivery of user data between eNBs. For example, the X2-U may provide specific sequence number information for user data transferred from a MeNB to an SeNB; information about successful in sequence delivery of PDCP PDUs to a UE **101** from an SeNB for user data; information of PDCP PDUs that were not delivered to a UE **101**; information about a current

minimum desired buffer size at the SeNB for transmitting to the UE user data; and the like. The X2-C may provide intra-LTE access mobility functionality, including context transfers from source to target eNBs, user plane transport control, etc.; load management functionality; as well as inter-cell interference coordination functionality.

In examples where the system 100 is a 5G or NR system, the interface 112 may be an Xn interface 112. The Xn interface is defined between two or more RAN nodes 111 (e.g., two or more gNBs and the like) that connect to 5GC 120, between a RAN node 111 (e.g., a gNB) connecting to 5GC 120 and an eNB, and/or between two eNBs connecting to 5GC 120. In some implementations, the Xn interface may include an Xn user plane (Xn-U) interface and an Xn control plane (Xn-C) interface. The Xn-U may provide non-guaranteed delivery of user plane PDUs and support/provide data forwarding and flow control functionality. The Xn-C may provide management and error handling functionality, functionality to manage the Xn-C interface; mobility support for UE 101 in a connected mode (e.g., CM-CONNECTED) including functionality to manage the UE mobility for connected mode between one or more RAN nodes 111. The mobility support may include context transfer from an old (source) serving RAN node 111 to new (target) serving RAN node 111; and control of user plane tunnels between old (source) serving RAN node 111 to new (target) serving RAN node 111. A protocol stack of the Xn-U may include a transport network layer built on Internet Protocol (IP) transport layer, and a GTP-U layer on top of a UDP and/or IP layer(s) to carry user plane PDUs. The Xn-C protocol stack may include an application layer signaling protocol (referred to as Xn Application Protocol (Xn-AP)) and a transport network layer that is built on SCTP. The SCTP may be on top of an IP layer, and may provide the guaranteed delivery of application layer messages. In the transport IP layer, point-to-point transmission is used to deliver the signaling PDUs. In other implementations, the Xn-U protocol stack and/or the Xn-C protocol stack may be same or similar to the user plane and/or control plane protocol stack(s) shown and described herein.

The RAN 110 is shown to be communicatively coupled to a core network-in this example, core network (CN) 120. The CN 120 may comprise a plurality of network elements 122, which are configured to offer various data and telecommunications services to customers/subscribers (e.g., users of UEs 101) who are connected to the CN 120 via the RAN 110. The components of the CN 120 may be implemented in one physical node or separate physical nodes including components to read and execute instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium). In some examples, NFV may be utilized to virtualize any or all of the above-described network node functions via executable instructions stored in one or more computer-readable storage mediums (described in further detail below). A logical instantiation of the CN 120 may be referred to as a network slice, and a logical instantiation of a portion of the CN 120 may be referred to as a network sub-slice. NFV architectures and infrastructures may be used to virtualize one or more network functions, alternatively performed by proprietary hardware, onto physical resources comprising a combination of industry-standard server hardware, storage hardware, or switches. In other words, NFV systems can be used to execute virtual or reconfigurable implementations of one or more EPC components/functions.

Generally, the application server 130 may be an element offering applications that use IP bearer resources with the

core network (e.g., UMTS PS domain, LTE PS data services, etc.). The application server 130 can also be configured to support one or more communication services (e.g., VoIP sessions, PTT sessions, group communication sessions, social networking services, etc.) for the UEs 101 via the EPC 120.

In some examples, the CN 120 may be a 5GC (referred to as “5GC 120” or the like), and the RAN 110 may be connected with the CN 120 via an NG interface 113. In some examples, the NG interface 113 may be split into two parts, an NG user plane (NG-U) interface 114, which carries traffic data between the RAN nodes 111 and a UPF, and the S1 control plane (NG-C) interface 115, which is a signaling interface between the RAN nodes 111 and AMFs.

In some examples, the CN 120 may be a 5G CN (referred to as “5GC 120” or the like), while in other examples, the CN 120 may be an EPC. Where CN 120 is an EPC (referred to as “EPC 120” or the like), the RAN 110 may be connected with the CN 120 via an S1 interface 113. In some examples, the S1 interface 113 may be split into two parts, an S1 user plane (S1-U) interface 114, which carries traffic data between the RAN nodes 111 and the S-GW, and the S1-MME interface 115, which is a signaling interface between the RAN nodes 111 and MMES.

FIG. 2 illustrates an example of infrastructure equipment 200 in accordance with various examples. The infrastructure equipment 200 (or “system 200”) may be implemented as a base station, radio head, RAN node such as the RAN nodes 111 and/or AP 106 shown and described previously, application server(s) 130, and/or any other element/device discussed herein. In other examples, the system 200 could be implemented in or by a UE.

The system 200 includes application circuitry 205, base-band circuitry 210, one or more radio front end modules (RFEMs) 215, memory circuitry 220, power management integrated circuitry (PMIC) 225, power tee circuitry 230, network controller circuitry 235, network interface connector 240, satellite positioning circuitry 245, and user interface 250. In some examples, the device 200 may include additional elements such as, for example, memory/storage, display, camera, sensor, or input/output (I/O) interface. In other examples, the components described below may be included in more than one device. For example, said circuitries may be separately included in more than one device for CRAN, vBBU, or other like implementations.

Application circuitry 205 includes circuitry such as, but not limited to one or more processors (or processor cores), cache memory, and one or more of low drop-out voltage regulators (LDOs), interrupt controllers, serial interfaces such as SPI, I2C or universal programmable serial interface module, real time clock (RTC), timer-counters including interval and watchdog timers, general purpose input/output (I/O or IO), memory card controllers such as Secure Digital (SD) MultiMediaCard (MMC) or similar, Universal Serial Bus (USB) interfaces, Mobile Industry Processor Interface (MIPI) interfaces and Joint Test Access Group (JTAG) test access ports. The processors (or cores) of the application circuitry 205 may be coupled with or may include memory/storage elements and may be configured to execute instructions stored in the memory/storage to enable various applications or operating systems to run on the system 200. In some implementations, the memory/storage elements may be on-chip memory circuitry, which may include any suitable volatile and/or non-volatile memory, such as DRAM, SRAM, EPROM, EEPROM, Flash memory, solid-state memory, and/or any other type of memory device technology, such as those discussed herein.

11

The processor(s) of application circuitry **205** may include, for example, one or more processor cores (CPUs), one or more application processors, one or more graphics processing units (GPUs), one or more reduced instruction set computing (RISC) processors, one or more Acorn RISC Machine (ARM) processors, one or more complex instruction set computing (CISC) processors, one or more digital signal processors (DSP), one or more FPGAs, one or more PLDs, one or more ASICs, one or more microprocessors or controllers, or any suitable combination thereof. In some examples, the application circuitry **205** may comprise, or may be, a special-purpose processor/controller to operate according to the various examples herein. As examples, the processor(s) of application circuitry **205** may include one or more Apple A-series processors, Intel Pentium®, Core®, or Xeon® processor(s); Advanced Micro Devices (AMD) Ryzen® processor(s), Accelerated Processing Units (APUs), or Epyc® processors; ARM-based processor(s) licensed from ARM Holdings, Ltd. Such as the ARM Cortex-A family of processors and the ThunderX2® provided by Cavium™, Inc.; a MIPS-based design from MIPS Technologies, Inc. Such as MIPS Warrior P-class processors; and/or the like. In some examples, the system **200** may not utilize application circuitry **205**, and instead may include a special-purpose processor/controller to process IP data received from an EPC or 5GC, for example.

In some implementations, the application circuitry **205** may include one or more hardware accelerators, which may be microprocessors, programmable processing devices, or the like. The one or more hardware accelerators may include, for example, computer vision (CV) and/or deep learning (DL) accelerators. As examples, the programmable processing devices may be one or more a field-programmable devices (FPDs) such as field-programmable gate arrays (FPGAs) and the like; programmable logic devices (PLDs) such as complex PLDs (CPLDs), high-capacity PLDs (HCPLDs), and the like; ASICs such as structured ASICs and the like; programmable SoCs (PSoCs); and the like. In such implementations, the circuitry of application circuitry **205** may comprise logic blocks or logic fabric, and other interconnected resources that may be programmed to perform various functions, such as the procedures, methods, functions, etc. Of the various examples discussed herein. In such examples, the circuitry of application circuitry **205** may include memory cells (e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), flash memory, static memory (e.g., static random access memory (SRAM), antifuses, etc.)) used to store logic blocks, logic fabric, data, etc. In look-up-tables (LUTs) and the like.

The baseband circuitry **210** may include circuitry and/or control logic configured to carry out various radio/network protocol and radio control functions that enable communication with one or more radio networks via the RF circuitry. In some examples, the baseband circuitry **210** may be implemented, for example, as a solder-down substrate including one or more integrated circuits, a single packaged integrated circuit soldered to a main circuit board or a multi-chip module containing two or more integrated circuits.

User interface circuitry **250** may include one or more user interfaces designed to enable user interaction with the system **200** or peripheral component interfaces designed to enable peripheral component interaction with the system **200**. User interfaces may include, but are not limited to, one or more physical or virtual buttons (e.g., a reset button), one or more indicators (e.g., light emitting diodes (LEDs)), a

12

physical keyboard or keypad, a mouse, a touchpad, a touch-screen, speakers or other audio emitting devices, microphones, a printer, a scanner, a headset, a display screen or display device, etc. Peripheral component interfaces may include, but are not limited to, a nonvolatile memory port, a universal serial bus (USB) port, an audio jack, a power supply interface, etc.

The radio front end modules (RFEMs) **215** may comprise a millimeter wave (mmWave) RFEM and one or more sub-mmWave radio frequency integrated circuits (RFICs). In some implementations, the one or more sub-mmWave RFICs may be physically separated from the mmWave RFEM. The RFICs may include connections to one or more antennas or antenna arrays, and the RFEM may be connected to multiple antennas. In alternative implementations, both mmWave and sub-mmWave radio functions may be implemented in the same physical RFEM **215**, which incorporates both mmWave antennas and sub-mmWave.

The memory circuitry **220** may include one or more of volatile memory including dynamic random access memory (DRAM) and/or synchronous dynamic random access memory (SDRAM), and nonvolatile memory (NVM) including high-speed electrically erasable memory (commonly referred to as Flash memory), phase change random access memory (PRAM), magnetoresistive random access memory (MRAM), etc., and may incorporate the three-dimensional (3D) cross-point (XPOINT) memories from Intel® and Micron®. Memory circuitry **220** may be implemented as one or more of solder down packaged integrated circuits, socketed memory modules and plug-in memory cards.

The PMIC **225** may include voltage regulators, surge protectors, power alarm detection circuitry, and one or more backup power sources such as a battery or capacitor. The power alarm detection circuitry may detect one or more of brown out (under-voltage) and surge (over-voltage) conditions. The power tee circuitry **230** may provide for electrical power drawn from a network cable to provide both power supply and data connectivity to the infrastructure equipment **200** using a single cable.

The network controller circuitry **235** may provide connectivity to a network using a standard network interface protocol such as Ethernet, Ethernet over GRE Tunnels, Ethernet over Multiprotocol Label Switching (MPLS), or some other suitable protocol. Network connectivity may be provided to/from the infrastructure equipment **200** via network interface connector **240** using a physical connection, which may be electrical (commonly referred to as a “copper interconnect”), optical, or wireless. The network controller circuitry **235** may include one or more dedicated processors and/or FPGAs to communicate using one or more of the aforementioned protocols. In some implementations, the network controller circuitry **235** may include multiple controllers to provide connectivity to other networks using the same or different protocols.

The positioning circuitry **245** includes circuitry to receive and decode signals transmitted/broadcasted by a positioning network of a global navigation satellite system (GNSS). Examples of navigation satellite constellations (or GNSS) include United States’ Global Positioning System (GPS), Russia’s Global Navigation System (GLONASS), the European Union’s Galileo system, China’s BeiDou Navigation Satellite System, a regional navigation system or GNSS augmentation system (e.g., Navigation with Indian Constellation (NAVIC), Japan’s Quasi-Zenith Satellite System (QZSS), France’s Doppler Orbitography and Radio-positioning Integrated by Satellite (DORIS), etc.), or the like.

13

The positioning circuitry **245** comprises various hardware elements (e.g., including hardware devices such as switches, filters, amplifiers, antenna elements, and the like to facilitate OTA communications) to communicate with components of a positioning network, such as navigation satellite constellation nodes. In some examples, the positioning circuitry **245** may include a Micro-Technology for Positioning, Navigation, and Timing (Micro-PNT) IC that uses a master timing clock to perform position tracking/estimation without GNSS assistance. The positioning circuitry **245** may also be part of, or interact with, the baseband circuitry **210** and/or RFEMs **215** to communicate with the nodes and components of the positioning network. The positioning circuitry **245** may also provide position data and/or time data to the application circuitry **205**, which may use the data to synchronize operations with various infrastructure (e.g., RAN nodes **111**, etc.), or the like.

The components shown by FIG. **2** may communicate with one another using interface circuitry, which may include any number of bus and/or interconnect (IX) technologies such as industry standard architecture (ISA), extended ISA (EISA), peripheral component interconnect (PCI), peripheral component interconnect extended (PCIx), PCI express (PCIe), or any number of other technologies. The bus/IX may be a proprietary bus, for example, used in a SoC based system. Other bus/IX systems may be included, such as an **12C** interface, an SPI interface, point to point interfaces, and a power bus, among others.

FIG. **3** illustrates an example of a platform **300** (or “device **300**”) in accordance with various examples. In some examples, the computer platform **300** may be suitable for use as UEs **101**, application servers **130**, and/or any other element/device discussed herein. The platform **300** may include any combinations of the components shown in the example. The components of platform **300** may be implemented as integrated circuits (ICs), portions thereof, discrete electronic devices, or other modules, logic, hardware, software, firmware, or a combination thereof adapted in the computer platform **300**, or as components otherwise incorporated within a chassis of a larger system. The block diagram of FIG. **3** is intended to show a high level view of components of the computer platform **300**. However, some of the components shown may be omitted, additional components may be present, and different arrangement of the components shown may occur in other implementations.

Application circuitry **305** includes circuitry such as, but not limited to one or more processors (or processor cores), cache memory, and one or more of LDOs, interrupt controllers, serial interfaces such as SPI, I2C or universal programmable serial interface module, RTC, timer-counters including interval and watchdog timers, general purpose I/O, memory card controllers such as SD MMC or similar, USB interfaces, MIPI interfaces, and JTAG test access ports. The processors (or cores) of the application circuitry **305** may be coupled with or may include memory/storage elements and may be configured to execute instructions stored in the memory/storage to enable various applications or operating systems to run on the system **300**. In some implementations, the memory/storage elements may be on-chip memory circuitry, which may include any suitable volatile and/or non-volatile memory, such as DRAM, SRAM, EPROM, EEPROM, Flash memory, solid-state memory, and/or any other type of memory device technology, such as those discussed herein.

The processor(s) of application circuitry **205** may include, for example, one or more processor cores, one or more application processors, one or more GPUs, one or more

14

RISC processors, one or more ARM processors, one or more CISC processors, one or more DSP, one or more FPGAs, one or more PLDs, one or more ASICs, one or more microprocessors or controllers, a multithreaded processor, an ultra-low voltage processor, an embedded processor, some other known processing element, or any suitable combination thereof. In some examples, the application circuitry **205** may comprise, or may be, a special-purpose processor/controller to operate according to the various examples herein.

As examples, the processor(s) of application circuitry **305** may include an Apple A-series processor. The processors of the application circuitry **305** may also be one or more of an Intel® Architecture Core™ based processor, such as a Quark™, an Atom™, an i3, an i5, an i7, or an MCU-class processor, or another such processor available from Intel® Corporation, Santa Clara, CA; Advanced Micro Devices (AMD) Ryzen® processor(s) or Accelerated Processing Units (APUs); Snapdragon™ processor(s) from Qualcomm® Technologies, Inc., Texas Instruments, Inc.® Open Multimedia Applications Platform (OMAP)™ processor(s); a MIPS-based design from MIPS Technologies, Inc. such as MIPS Warrior M-class, Warrior I-class, and Warrior P-class processors; an ARM-based design licensed from ARM Holdings, Ltd., such as the ARM Cortex-A, Cortex-R, and Cortex-M family of processors; or the like. In some implementations, the application circuitry **305** may be a part of a system on a chip (SoC) in which the application circuitry **305** and other components are formed into a single integrated circuit.

Additionally or alternatively, application circuitry **305** may include circuitry such as, but not limited to, one or more a field-programmable devices (FPDs) such as FPGAs and the like; programmable logic devices (PLDs) such as complex PLDs (CPLDs), high-capacity PLDs (HCPLDs), and the like; ASICs such as structured ASICs and the like; programmable SoCs (PSoCs); and the like. In such examples, the circuitry of application circuitry **305** may comprise logic blocks or logic fabric, and other interconnected resources that may be programmed to perform various functions, such as the procedures, methods, functions, etc. Of the various examples discussed herein. In such examples, the circuitry of application circuitry **305** may include memory cells (e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), flash memory, static memory (e.g., static random access memory (SRAM), anti-fuses, etc.)) used to store logic blocks, logic fabric, data, etc. In look-up tables (LUTs) and the like.

The baseband circuitry **310** includes circuitry and/or control logic configured to carry out various radio/network protocol and radio control functions that enable communication with one or more radio networks via the RF circuitry. In some examples, the baseband circuitry **310** may be implemented, for example, as a solder-down substrate including one or more integrated circuits, a single packaged integrated circuit soldered to a main circuit board or a multi-chip module containing two or more integrated circuits.

The RFEMs **315** may comprise a millimeter wave (mm-Wave) RFEM and one or more sub-mmWave radio frequency integrated circuits (RFICs). In some implementations, the one or more sub-mmWave RFICs may be physically separated from the mmWave RFEM. The RFICs may include connections to one or more antennas or antenna arrays, and the RFEM may be connected to multiple antennas. In alternative implementations, both mmWave and sub-mmWave radio functions may be implemented in the

15

same physical RFEM **315**, which incorporates both mmWave antennas and sub-mmWave.

The memory circuitry **320** may include any number and type of memory devices used to provide for a given amount of system memory. As examples, the memory circuitry **320** may include one or more of volatile memory including random access memory (RAM), dynamic RAM (DRAM) and/or synchronous dynamic RAM (SDRAM), and nonvolatile memory (NVM) including high-speed electrically erasable memory (commonly referred to as Flash memory), phase change random access memory (PRAM), magnetoresistive random access memory (MRAM), etc. The memory circuitry **320** may be developed in accordance with a Joint Electron Devices Engineering Council (JEDEC) low power double data rate (LPDDR)-based design, such as LPDDR2, LPDDR3, LPDDR4, or the like. Memory circuitry **320** may be implemented as one or more of solder down packaged integrated circuits, single die package (SDP), dual die package (DDP) or quad die package (Q17P), socketed memory modules, dual inline memory modules (DIMMs) including microDIMMs or MiniDIMMs, and/or soldered onto a motherboard via a ball grid array (BGA). In low power implementations, the memory circuitry **320** may be on-die memory or registers associated with the application circuitry **305**. To provide for persistent storage of information such as data, applications, operating systems and so forth, memory circuitry **320** may include one or more mass storage devices, which may include, inter alia, a solid state disk drive (SSDD), hard disk drive (HDD), a micro HDD, resistance change memories, phase change memories, holographic memories, or chemical memories, among others. For example, the computer platform **300** may incorporate the three-dimensional (3D) cross-point (XPOINT) memories from Intel® and Micron®.

Removable memory circuitry **323** may include devices, circuitry, enclosures/housings, ports or receptacles, etc. Used to couple portable data storage devices with the platform **300**. These portable data storage devices may be used for mass storage purposes, and may include, for example, flash memory cards (e.g., Secure Digital (SD) cards, microSD cards, xD picture cards, and the like), and USB flash drives, optical discs, external HDDs, and the like.

The platform **300** may also include interface circuitry (not shown) that is used to connect external devices with the platform **300**. The external devices connected to the platform **300** via the interface circuitry include sensor circuitry **321** and electro-mechanical components (EMCs) **322**, as well as removable memory devices coupled to removable memory circuitry **323**.

The sensor circuitry **321** include devices, modules, or subsystems whose purpose is to detect events or changes in its environment and send the information (sensor data) about the detected events to some other a device, module, subsystem, etc. Examples of such sensors include, inter alia, inertia measurement units (IMUs) comprising accelerometers, gyroscopes, and/or magnetometers; microelectromechanical systems (MEMS) or nanoelectromechanical systems (NEMS) comprising 3-axis accelerometers, 3-axis gyroscopes, and/or magnetometers; level sensors; flow sensors; temperature sensors (e.g., thermistors); pressure sensors; barometric pressure sensors; gravimeters; altimeters; image capture devices (e.g., cameras or lensless apertures); light detection and ranging (LiDAR) sensors; proximity sensors (e.g., infrared radiation detector and the like), depth sensors, ambient light sensors, ultrasonic transceivers; microphones or other like audio capture devices; etc.

16

EMCs **322** include devices, modules, or subsystems whose purpose is to enable platform **300** to change its state, position, and/or orientation, or move or control a mechanism or (sub)system. Additionally, EMCs **322** may be configured to generate and send messages/signaling to other components of the platform **300** to indicate a current state of the EMCs **322**. Examples of the EMCs **322** include one or more power switches, relays including electromechanical relays (EMRs) and/or solid state relays (SSRs), actuators (e.g., valve actuators, etc.), an audible sound generator, a visual warning device, motors (e.g., DC motors, stepper motors, etc.), wheels, thrusters, propellers, claws, clamps, hooks, and/or other like electro-mechanical components. In some examples, platform **300** is configured to operate one or more EMCs **322** based on one or more captured events and/or instructions or control signals received from a service provider and/or various clients.

In some implementations, the interface circuitry may connect the platform **300** with positioning circuitry **345**. The positioning circuitry **345** includes circuitry to receive and decode signals transmitted/broadcasted by a positioning network of a GNSS. Examples of navigation satellite constellations (or GNSS) include United States' GPS, Russia's GLONASS, the European Union's Galileo system, China's BeiDou Navigation Satellite System, a regional navigation system or GNSS augmentation system (e.g., NAVIC), Japan's QZSS, France's DORIS, etc.), or the like. The positioning circuitry **345** comprises various hardware elements (e.g., including hardware devices such as switches, filters, amplifiers, antenna elements, and the like to facilitate OTA communications) to communicate with components of a positioning network, such as navigation satellite constellation nodes. In some examples, the positioning circuitry **345** may include a Micro-PNT IC that uses a master timing clock to perform position tracking/estimation without GNSS assistance. The positioning circuitry **345** may also be part of, or interact with, the baseband circuitry **210** and/or RFEMs **315** to communicate with the nodes and components of the positioning network. The positioning circuitry **345** may also provide position data and/or time data to the application circuitry **305**, which may use the data to synchronize operations with various infrastructure (e.g., radio base stations), for turn-by-turn navigation applications, or the like.

In some implementations, the interface circuitry may connect the platform **300** with Near-Field Communication (NFC) circuitry **340**. NFC circuitry **340** is configured to provide contactless, short-range communications based on radio frequency identification (RFID) standards, wherein magnetic field induction is used to enable communication between NFC circuitry **340** and NFC-enabled devices external to the platform **300** (e.g., an "NFC touchpoint"). NFC circuitry **340** comprises an NFC controller coupled with an antenna element and a processor coupled with the NFC controller. The NFC controller may be a chip/IC providing NFC functionalities to the NFC circuitry **340** by executing NFC controller firmware and an NFC stack. The NFC stack may be executed by the processor to control the NFC controller, and the NFC controller firmware may be executed by the NFC controller to control the antenna element to emit short-range RF signals. The RF signals may power a passive NFC tag (e.g., a microchip embedded in a sticker or wristband) to transmit stored data to the NFC circuitry **340**, or initiate data transfer between the NFC circuitry **340** and another active NFC device (e.g., a smartphone or an NFC-enabled POS terminal) that is proximate to the platform **300**.

The driver circuitry **346** may include software and hardware elements that operate to control particular devices that are embedded in the platform **300**, attached to the platform **300**, or otherwise communicatively coupled with the platform **300**. The driver circuitry **346** may include individual drivers allowing other components of the platform **300** to interact with or control various input/output (I/O) devices that may be present within, or connected to, the platform **300**. For example, driver circuitry **346** may include a display driver to control and allow access to a display device, a touchscreen driver to control and allow access to a touchscreen interface of the platform **300**, sensor drivers to obtain sensor readings of sensor circuitry **321** and control and allow access to sensor circuitry **321**, EMC drivers to obtain actuator positions of the EMCs **322** and/or control and allow access to the EMCs **322**, a camera driver to control and allow access to an embedded image capture device, audio drivers to control and allow access to one or more audio devices.

The power management integrated circuitry (PMIC) **325** (also referred to as “power management circuitry **325**”) may manage power provided to various components of the platform **300**. In particular, with respect to the baseband circuitry **310**, the PMIC **325** may control power-source selection, voltage scaling, battery charging, or DC-to-DC conversion. The PMIC **325** may often be included when the platform **300** is capable of being powered by a battery **330**, for example, when the device is included in a UE **101**.

A battery **330** may power the platform **300**, although in some examples the platform **300** may be mounted deployed in a fixed location, and may have a power supply coupled to an electrical grid. The battery **330** may be a lithium ion battery, a metal-air battery, such as a zinc-air battery, an aluminum-air battery, a lithium-air battery, and the like. In some implementations, such as in V2X applications, the battery **330** may be a typical lead-acid automotive battery.

In some implementations, the battery **330** may be a “smart battery,” which includes or is coupled with a Battery Management System (BMS) or battery monitoring integrated circuitry. The BMS may be included in the platform **300** to track the state of charge (SoCh) of the battery **330**. The BMS may be used to monitor other parameters of the battery **330** to provide failure predictions, such as the state of health (SoH) and the state of function (SoF) of the battery **330**. The BMS may communicate the information of the battery **330** to the application circuitry **305** or other components of the platform **300**. The BMS may also include an analog-to-digital (ADC) convertor that allows the application circuitry **305** to directly monitor the voltage of the battery **330** or the current flow from the battery **330**. The battery parameters may be used to determine actions that the platform **300** may perform, such as transmission frequency, network operation, sensing frequency, and the like.

A power block, or other power supply coupled to an electrical grid may be coupled with the BMS to charge the battery **330**. In some examples, the power block **XS30** may be replaced with a wireless power receiver to obtain the power wirelessly, for example, through a loop antenna in the computer platform **300**. In these examples, a wireless battery charging circuit may be included in the BMS. The specific charging circuits chosen may depend on the size of the battery **330**, and thus, the current required. The charging may be performed using the Airfuel standard promulgated by the Airfuel Alliance, the Qi wireless charging standard promulgated by the Wireless Power Consortium, or the Rezence charging standard promulgated by the Alliance for Wireless Power, among others.

User interface circuitry **350** includes various input/output (I/O) devices present within, or connected to, the platform **300**, and includes one or more user interfaces designed to enable user interaction with the platform **300** and/or peripheral component interfaces designed to enable peripheral component interaction with the platform **300**. The user interface circuitry **350** includes input device circuitry and output device circuitry. Input device circuitry includes any physical or virtual means for accepting an input including, inter alia, one or more physical or virtual buttons (e.g., a reset button), a physical keyboard, keypad, mouse, touchpad, touchscreen, microphones, scanner, headset, and/or the like. The output device circuitry includes any physical or virtual means for showing information or otherwise conveying information, such as sensor readings, actuator position (s), or other like information. Output device circuitry may include any number and/or combinations of audio or visual display, including, inter alia, one or more simple visual outputs/indicators (e.g., binary status indicators (e.g., light emitting diodes (LEDs))) and multi-character visual outputs, or more complex outputs such as display devices or touchscreens (e.g., Liquid Chrystal Displays (LCD), LED displays, quantum dot displays, projectors, etc.), with the output of characters, graphics, multimedia objects, and the like being generated or produced from the operation of the platform **300**. The output device circuitry may also include speakers or other audio emitting devices, printer(s), and/or the like. In some examples, the sensor circuitry **321** may be used as the input device circuitry (e.g., an image capture device, motion capture device, or the like) and one or more EMCs may be used as the output device circuitry (e.g., an actuator to provide haptic feedback or the like). In another example, NFC circuitry comprising an NFC controller coupled with an antenna element and a processing device may be included to read electronic tags and/or connect with another NFC-enabled device. Peripheral component interfaces may include, but are not limited to, a non-volatile memory port, a USB port, an audio jack, a power supply interface, etc.

Although not shown, the components of platform **300** may communicate with one another using a suitable bus or interconnect (IX) technology, which may include any number of technologies, including ISA, EISA, PCI, PCIX, PCIe, a Time-Trigger Protocol (TTP) system, a FlexRay system, or any number of other technologies. The bus/IX may be a proprietary bus/IX, for example, used in a SoC based system. Other bus/IX systems may be included, such as an I2C interface, an SPI interface, point-to-point interfaces, and a power bus, among others.

FIG. 4 illustrates various protocol functions that may be implemented in a wireless communication device according to various examples. In particular, FIG. 4 includes an arrangement **400** showing interconnections between various protocol layers/entities. The following description of FIG. 4 is provided for various protocol layers/entities that operate in conjunction with the 5G/NR system standards and LTE system standards, but some or all of the aspects of FIG. 4 may be applicable to other wireless communication network systems as well.

The protocol layers of arrangement **400** may include one or more of PHY **410**, MAC **420**, RLC **430**, PDCP **440**, SDAP **447**, RRC **455**, and NAS layer **457**, in addition to other higher layer functions not illustrated. The protocol layers may include one or more service access points (e.g., items **459**, **456**, **450**, **449**, **445**, **435**, **425**, and **415** in FIG. 4) that may provide communication between two or more protocol layers.

The PHY 410 may transmit and receive physical layer signals 405 that may be received from or transmitted to one or more other communication devices. The physical layer signals 405 may comprise one or more physical channels, such as those discussed herein. The PHY 410 may further perform link adaptation or adaptive modulation and coding (AMC), power control, cell search (e.g., for initial synchronization and handover purposes), and other measurements used by higher layers, such as the RRC 455. The PHY 410 may still further perform error detection on the transport channels, forward error correction (FEC) coding/decoding of the transport channels, modulation/demodulation of physical channels, interleaving, rate matching, mapping onto physical channels, and MIMO antenna processing. In some examples, an instance of PHY 410 may process requests from and provide indications to an instance of MAC 420 via one or more PHY-SAP 415. According to some examples, requests and indications communicated via PHY-SAP 415 may comprise one or more transport channels.

Instance(s) of MAC 420 may process requests from, and provide indications to, an instance of RLC 430 via one or more MAC-SAPs 425. These requests and indications communicated via the MAC-SAP 425 may comprise one or more logical channels. The MAC 420 may perform mapping between the logical channels and transport channels, multiplexing of MAC SDUs from one or more logical channels onto TBs to be delivered to PHY 410 via the transport channels, de-multiplexing MAC SDUs to one or more logical channels from TBs delivered from the PHY 410 via transport channels, multiplexing MAC SDUs onto TBs, scheduling information reporting, error correction through HARQ, and logical channel prioritization.

Instance(s) of RLC 430 may process requests from and provide indications to an instance of PDCP 440 via one or more radio link control service access points (RLC-SAP) 435. These requests and indications communicated via RLC-SAP 435 may comprise one or more RLC channels. The RLC 430 may operate in a plurality of modes of operation, including: Transparent Mode (TM), Unacknowledged Mode (UM), and Acknowledged Mode (AM). The RLC 430 may execute transfer of upper layer protocol data units (PDUs), error correction through automatic repeat request (ARQ) for AM data transfers, and concatenation, segmentation and reassembly of RLC SDUs for UM and AM data transfers. The RLC 430 may also execute re-segmentation of RLC data PDUs for AM data transfers, reorder RLC data PDUs for UM and AM data transfers, detect duplicate data for UM and AM data transfers, discard RLC SDUs for UM and AM data transfers, detect protocol errors for AM data transfers, and perform RLC re-establishment.

Instance(s) of PDCP 440 may process requests from and provide indications to instance(s) of RRC 455 and/or instance(s) of SDAP 447 via one or more packet data convergence protocol service access points (PDCP-SAP) 445. These requests and indications communicated via PDCP-SAP 445 may comprise one or more radio bearers. The PDCP 440 may execute header compression and decompression of IP data, maintain PDCP Sequence Numbers (SNs), perform in-sequence delivery of upper layer PDUs at re-establishment of lower layers, eliminate duplicates of lower layer SDUs at re-establishment of lower layers for radio bearers mapped on RLC AM, cipher and decipher control plane data, perform integrity protection and integrity verification of control plane data, control timer-

based discard of data, and perform security operations (e.g., ciphering, deciphering, integrity protection, integrity verification, etc.).

Instance(s) of SDAP 447 may process requests from and provide indications to one or more higher layer protocol entities via one or more SDAP-SAP 449. These requests and indications communicated via SDAP-SAP 449 may comprise one or more QoS flows. The SDAP 447 may map QoS flows to DRBs, and vice versa, and may also mark QFIs in DL and UL packets. A single SDAP entity 447 may be configured for an individual PDU session. In the UL direction, the NG-RAN 110 may control the mapping of QoS Flows to DRB(s) in two different ways, reflective mapping or explicit mapping. For reflective mapping, the SDAP 447 of a UE 101 may monitor the QFIs of the DL packets for each DRB, and may apply the same mapping for packets flowing in the UL direction. For a DRB, the SDAP 447 of the UE 101 may map the UL packets belonging to the QoS flows(s) corresponding to the QoS flow ID(s) and PDU session observed in the DL packets for that DRB. To enable reflective mapping, the NG-RAN may mark DL packets over the Uu interface with a QoS flow ID. The explicit mapping may involve the RRC 455 configuring the SDAP 447 with an explicit QoS flow to DRB mapping rule, which may be stored and followed by the SDAP 447. In some examples, the SDAP 447 may only be used in NR implementations and may not be used in LTE implementations.

The RRC 455 may configure, via one or more management service access points (M-SAP), aspects of one or more protocol layers, which may include one or more instances of PHY 410, MAC 420, RLC 430, PDCP 440 and SDAP 447. In some examples, an instance of RRC 455 may process requests from and provide indications to one or more NAS entities 457 via one or more RRC-SAPs 456. The main services and functions of the RRC 455 may include broadcast of system information (e.g., included in MIBs or SIBs related to the NAS), broadcast of system information related to the access stratum (AS), paging, establishment, maintenance and release of an RRC connection between the UE 101 and RAN 110 (e.g., RRC connection paging, RRC connection establishment, RRC connection modification, and RRC connection release), establishment, configuration, maintenance and release of point to point Radio Bearers, security functions including key management, inter-RAT mobility, and measurement configuration for UE measurement reporting. The MIBs and SIBs may comprise one or more IEs, which may each comprise individual data fields or data structures.

The NAS 457 may form the highest stratum of the control plane between the UE 101 and the AMF. The NAS 457 may support the mobility of the UEs 101 and the session management procedures to establish and maintain IP connectivity between the UE 101 and a P-GW in LTE systems.

According to various examples, one or more protocol entities of arrangement 400 may be implemented in UEs 101, RAN nodes 111, AMF (e.g., in NR implementations) or MME (e.g., in LTE implementations), UPF (e.g., in NR implementations) or S-GW and P-GW (e.g., in LTE implementations), or the like to be used for control plane or user plane communications protocol stack between the aforementioned devices. In such examples, one or more protocol entities that may be implemented in one or more of UE 101, gNB 111, AMF, etc. May communicate with a respective peer protocol entity that may be implemented in or on another device using the services of respective lower layer protocol entities to perform such communication. In some examples, a gNB-CU of the gNB 111 may host the RRC 455,

SDAP 447, and PDCP 440 of the gNB that controls the operation of one or more gNB-DUs, and the gNB-DUs of the gNB 111 may each host the RLC 430, MAC 420, and PHY 410 of the gNB 111.

In a first example, a control plane protocol stack may comprise, in order from highest layer to lowest layer, NAS 457, RRC 455, PDCP 440, RLC 430, MAC 420, and PHY 410. In this example, upper layers 460 may be built on top of the NAS 457, which includes an IP layer 461, an SCTP 462, and an application layer signaling protocol (AP) 463.

In NR implementations, the AP 463 may be an NG application protocol layer (NGAP or NG-AP) 463 for the NG interface 113 defined between the NG-RAN node 111 and the AMF, or the AP 463 may be an Xn application protocol layer (XnAP or Xn-AP) 463 for the Xn interface 112 that is defined between two or more RAN nodes 111.

The NG-AP 463 may support the functions of the NG interface 113 and may comprise Elementary Procedures (EPs). An NG-AP EP may be a unit of interaction between the NG-RAN node 111 and the AMF. The NG-AP 463 services may comprise two groups: UE-associated services (e.g., services related to a UE 101) and non-UE-associated services (e.g., services related to the whole NG interface instance between the NG-RAN node 111 and AMF). These services may include functions including, but not limited to: a paging function for the sending of paging requests to NG-RAN nodes 111 involved in a particular paging area; a UE context management function for allowing the AMF to establish, modify, and/or release a UE context in the AMF and the NG-RAN node 111; a mobility function for UEs 101 in ECM-CONNECTED mode for intra-system HOs to support mobility within NG-RAN and inter-system HOs to support mobility from/to EPS systems; a NAS Signaling Transport function for transporting or rerouting NAS messages between UE 101 and AMF; a NAS node selection function for determining an association between the AMF and the UE 101; NG interface management function(s) for setting up the NG interface and monitoring for errors over the NG interface; a warning message transmission function for providing means to transfer warning messages via NG interface or cancel ongoing broadcast of warning messages; a Configuration Transfer function for requesting and transferring of RAN configuration information (e.g., SON information, performance measurement (PM) data, etc.) between two RAN nodes 111 via CN 120; and/or other like functions.

The XnAP 463 may support the functions of the Xn interface 112 and may comprise XnAP basic mobility procedures and XnAP global procedures. The XnAP basic mobility procedures may comprise procedures used to handle UE mobility within the NG RAN 111 (or E-UTRAN), such as handover preparation and cancellation procedures, SN Status Transfer procedures, UE context retrieval and UE context release procedures, RAN paging procedures, dual connectivity related procedures, and the like. The XnAP global procedures may comprise procedures that are not related to a specific UE 101, such as Xn interface setup and reset procedures, NG-RAN update procedures, cell activation procedures, and the like.

In LTE implementations, the AP 463 may be an S1 Application Protocol layer (S1-AP) 463 for the S1 interface 113 defined between an E-UTRAN node 111 and an MME, or the AP 463 may be an X2 application protocol layer (X2AP or X2-AP) 463 for the X2 interface 112 that is defined between two or more E-UTRAN nodes 111.

The S1 Application Protocol layer (S1-AP) 463 may support the functions of the S1 interface, and similar to the NG-AP discussed previously, the S1-AP may comprise

S1-AP EPs. An S1-AP EP may be a unit of interaction between the E-UTRAN node 111 and an MME within an LTE CN 120. The S1-AP 463 services may comprise two groups: UE-associated services and non UE-associated services. These services perform functions including, but not limited to: E-UTRAN Radio Access Bearer (E-RAB) management, UE capability indication, mobility, NAS signaling transport, RAN Information Management (RIM), and configuration transfer.

The X2AP 463 may support the functions of the X2 interface 112 and may comprise X2AP basic mobility procedures and X2AP global procedures. The X2AP basic mobility procedures may comprise procedures used to handle UE mobility within the E-UTRAN 120, such as handover preparation and cancellation procedures, SN Status Transfer procedures, UE context retrieval and UE context release procedures, RAN paging procedures, dual connectivity related procedures, and the like. The X2AP global procedures may comprise procedures that are not related to a specific UE 101, such as X2 interface setup and reset procedures, load indication procedures, error indication procedures, cell activation procedures, and the like.

The SCTP layer (alternatively referred to as the SCTP/IP layer) 462 may provide guaranteed delivery of application layer messages (e.g., NGAP or XnAP messages in NR implementations, or S1-AP or X2AP messages in LTE implementations). The SCTP 462 may ensure reliable delivery of signaling messages between the RAN node 111 and the AMF/MME based, in part, on the IP protocol, supported by the IP 461. The Internet Protocol layer (IP) 461 may be used to perform packet addressing and routing functionality. In some implementations the IP layer 461 may use point-to-point transmission to deliver and convey PDUs. In this regard, the RAN node 111 may comprise L2 and L1 layer communication links (e.g., wired or wireless) with the MME/AMF to exchange information.

In a second example, a user plane protocol stack may comprise, in order from highest layer to lowest layer, SDAP 447, PDCP 440, RLC 430, MAC 420, and PHY 410. The user plane protocol stack may be used for communication between the UE 101, the RAN node 111, and UPF in NR implementations or an S-GW and P-GW in LTE implementations. In this example, upper layers 451 may be built on top of the SDAP 447, and may include a user datagram protocol (UDP) and IP security layer (UDP/IP) 452, a General Packet Radio Service (GPRS) Tunneling Protocol for the user plane layer (GTP-U) 453, and a User Plane PDU layer (UP PDU) 463.

The transport network layer 454 (also referred to as a "transport layer") may be built on IP transport, and the GTP-U 453 may be used on top of the UDP/IP layer 452 (comprising a UDP layer and IP layer) to carry user plane PDUs (UP-PDUs). The IP layer (also referred to as the "Internet layer") may be used to perform packet addressing and routing functionality. The IP layer may assign IP addresses to user data packets in any of IPv4, IPv6, or PPP formats, for example.

The GTP-U 453 may be used for carrying user data within the GPRS core network and between the radio access network and the core network. The user data transported can be packets in any of IPv4, IPv6, or PPP formats, for example. The UDP/IP 452 may provide checksums for data integrity, port numbers for addressing different functions at the source and destination, and encryption and authentication on the selected data flows. The RAN node 111 and the S-GW may utilize an S1-U interface to exchange user plane data via a protocol stack comprising an L1 layer (e.g., PHY

410), an L2 layer (e.g., MAC 420, RLC 430, PDCP 440, and/or SDAP 447), the UDP/IP layer 452, and the GTP-U 453. The S-GW and the P-GW may utilize an S5/S8a interface to exchange user plane data via a protocol stack comprising an L1 layer, an L2 layer, the UDP/IP layer 452, and the GTP-U 453. As discussed previously, NAS protocols may support the mobility of the UE 101 and the session management procedures to establish and maintain IP connectivity between the UE 101 and the P-GW.

Moreover, although not shown by FIG. 4, an application layer may be present above the AP 463 and/or the transport network layer 454. The application layer may be a layer in which a user of the UE 101, RAN node 111, or other network element interacts with software applications being executed, for example, by application circuitry 205 or application circuitry 305, respectively. The application layer may also provide one or more interfaces for software applications to interact with communications systems of the UE 101 or RAN node 111, such as the baseband circuitry 210, 310. In some implementations the IP layer and/or the application layer may provide the same or similar functionality as layers 5-7, or portions thereof, of the Open Systems Interconnection (OSI) model (e.g., OSI Layer 7—the application layer, OSI Layer 6—the presentation layer, and OSI Layer 5—the session layer).

As noted above, the 5G NR standard allows for a network to configure a UE (e.g., the UE 101 or device 300) to operate on a subset of the total channel bandwidth known as a bandwidth part (BWP). Under the standard, a UE can be configured with up to four BWPs in the uplink and/or downlink, with only one BWP being active in the uplink and downlink at a given time. Each BWP can be configured (e.g., through RRC messages) with a particular frequency location, bandwidth size, numerology (μ) (e.g., subcarrier spacing (SCS), symbol duration, and cyclic prefix (CP) length), and control resource set (CORESET). In this way, a UE can be configured with multiple BWPs each serving different purposes. For example, one BWP may have a reduced bandwidth size (e.g., to reduce UE power consumption or to support a UE that is not capable of operating on the full carrier bandwidth), while another BWP may have a larger bandwidth size (e.g., to support high rate data transfer), and yet another BWP may have other numerologies or parameters to allow for compatibility with other systems or networks (e.g., a LTE network).

To reduce UE complexity, one or more bandwidth restrictions can be applied to BWPs that eliminate the need for RF tuning in certain situations. For example, under the 5G NR standard, a DL BWP, such as a first active BWP or a UE-specific RRC-configured BWP, can include the bandwidths of the Synchronization Signal Block (SSB) and Control Resource Set (CORESET) #0 (if present) for the PCell/PSCell, and the bandwidth of the SSB for the SCell (if present). In this way, the UE can perform RRM activities, such as intra-frequency and serving cell measurements, SCell activation procedures, and PSCell addition, among others, without modification of the RF front end.

In some implementations, a UE may support BWP operation without the above-noted bandwidth restrictions. For example, a UE may be capable of receiving the SSB of a PCell, PSCell, and/or SCell outside of the DL BWP of the cell. Such a capability can be indicated by the UE (e.g., in a UE capability message) through use of the bwp-WithoutRestriction parameter defined in 3GPP technical specification (TS) 38.306 and shown in the following table. However, different UEs may implement BWP operation without bandwidth restrictions in different ways, making it difficult

for the UE and the network to coordinate RRM activities outside of the active DL BWP.

Definitions for parameters	Per	M	FDD-TDD DIFF	FR1-FR2 DIFF
bwp-WithoutRestriction	Band	No	N/A	N/A
Indicates support of BWP operation without bandwidth restriction. The Bandwidth restriction in terms of DL BWP for PCell and PSCell means that the bandwidth of a UE-specific RRC configured DL BWP may not include the bandwidth of CORESET #0 (if configured) and SSB. For SCell(s), it means that the bandwidth of DL BWP may not include SSB.				

The technology described here enables performance of RRM activities outside of an active DL BWP when a UE indicates support for BWP operation without bandwidth restriction. In particular, techniques for configuring a UE to receive SSBs outside of an active DL BWP are described. In addition, ways in which a network can interpret an indication of support for BWP operation without bandwidth restriction, including whether a UE requires a measurement gap or interruption(s) for RF adjustment, are provided. In this manner, a network and a UE can coordinate RRM activities including intra-frequency and serving cell measurements, SCell activation procedures, and PSCell addition, among others, outside of an active DL BWP, thereby improving RRM flexibility and performance.

In accordance with an aspect of the present disclosure, if a UE indicates support for BWP operation without bandwidth restriction, then the UE can be configured to expand its operating bandwidth to perform measurements on a SSB (e.g., a target intra-frequency or serving cell SSB) that is outside any configured DL BWP. For example, referring to FIG. 5A, a diagram 500 is shown in which a UE is operating on an active BWP 502. In this example, the UE determines that a target SSB 504 is outside the active BWP 502 (e.g., based on an indication from the network, such as an indication of the center frequency of target SSB or another frequency or frequency range associated with the target SSB). If the UE supports BWP operation without bandwidth restriction for this frequency layer or band (e.g., the frequency layer or band that includes the SSB 504), then the UE can adjust one or more RF communication parameters (e.g., parameters of the RF chain for the active BWP) to operate on an adjusted operating bandwidth (BW) 506 that includes both the active BWP 502 and a bandwidth (e.g., frequency range) of the SSB 504. In this example, the network (e.g., a RAN node 111 or system 200) can assume that the UE can receive the SSB 504 and perform SSB measurements without any measurement gap based at least in part on the indication that the UE supports BWP operation without bandwidth restriction. In this way, the UE can perform intra-frequency or serving cell measurements (among other RRM activities) on the SSB 504 outside the active BWP 502 without a measurement gap when the UE indicates support for BWP operation without restriction.

In some examples, instead of expanding the operating bandwidth of an active RF chain, a UE can use a spare RF chain to create an additional operating bandwidth to cover the SSB. For example, referring to FIG. 5B, a diagram 550 is shown in which a UE is operating on an active BWP 552. In this example, the UE determines that a target SSB 554 is

outside the active BWP **552**. If the UE supports BWP operation without bandwidth restriction, then the UE can adjust one or more RF communication parameters (e.g., configure a spare RF chain) to operate on both a first operating bandwidth **556** and a second operating bandwidth **558** separated by a gap **560**, in which the first operating bandwidth **556** includes the bandwidth of the active BWP **552**, and the second operating bandwidth **558** includes a bandwidth of the SSB **554**. In this example, the network can assume that the UE can receive the SSB **554** and perform SSB measurements without any measurement gap based at least in part on the indication that the UE supports BWP operation without bandwidth restriction. In this way, the UE can measure the SSB **554** outside the active BWP **552** without needing to operate in the bandwidth defined by the gap **560**.

In general, whether the UE expands the operating bandwidth of an active RF chain to cover the SSB (as shown in FIG. **5A**) or uses a spare RF chain to create an additional operating bandwidth that covers the SSB (as shown in FIG. **5B**) can depend on the particular UE implementation and capabilities. In some examples, the UE can choose to expand the operating bandwidth of an active RF chain to cover a target SSB when the separation between the active BWP and the target SSB in the frequency domain is below a particular threshold (e.g., below 40-100 MHz). On the other hand, when the separation between the active BWP and the target SSB in the frequency domain is above a threshold (e.g., above 100 MHz), then the UE can choose to use a spare RF chain to create an additional operating bandwidth that covers the target SSB. To facilitate this decision, the network can indicate to the UE one or more parameters of the target SSB, such as the center frequency, bandwidth, and the like. In some examples, standard availability and scheduling restrictions defined in the relevant technical specifications (e.g., the 5G technical specifications) can apply when performing measurements on the target SSB.

In some examples, it may be beneficial for a UE to increase its operating bandwidth only during a measurement occasion for a target SSB in order to reduce its power consumption outside of the measurement occasion. In this configuration, the UE may need time before and after the SSB measurement occasion to adjust its operating bandwidth. Accordingly, in some examples, if a UE indicates support for BWP operation without bandwidth restriction, then the UE can be configured to expand its operating bandwidth upon occurrence of a measurement occasion for a SSB that is outside any configured DL BWP, and an interruption period can be included before and after the SSB measurement occasion. For example, referring to FIG. **6A**, a diagram **600** is shown in which a UE is operating on an active BWP **602**. In this example, the UE determines that a target SSB **604** having a measurement occasion **606** is outside the active BWP **602**. If the UE indicates support for BWP operation without bandwidth restriction, then the UE can adjust one or more RF communication parameters to expand from a first operating bandwidth **608** to a second operating bandwidth **610** upon occurrence of the measurement occasion **606** for the SSB **604**. The first operating bandwidth **608** can correspond to the active BWP **602**, and the second operating bandwidth **610** can include both the active BWP **602** and a bandwidth of the SSB **604**. Once the SSB measurement occasion **606** ends, the UE can revert its operating bandwidth to the first operating bandwidth **608**. In this example, the network can cause interruptions **612**, **614** before and after the SSB measurement occasion **606** to allow the UE sufficient time for RF adjustment based at least in

part on the indication that the UE supports BWP operation without bandwidth restriction. In addition, the network can assume that the UE can receive the SSB **604** and perform SSB measurements without any measurement gap based at least in part on the indication that the UE supports BWP operation without bandwidth restriction.

In some examples, a UE can configure a spare RF chain to create an additional operating bandwidth to cover the SSB during the SSB measurement occasion. For example, referring to FIG. **6B**, a diagram **650** is shown in which a UE is operating on an active BWP **652**. In this example, the UE determines that a target SSB **654** having a measurement occasion **656** is outside the active BWP **652**. Initially, the UE operates on a first operating bandwidth **658** corresponding to the active BWP **652**. Upon occurrence of the SSB measurement occasion **656**, the UE can activate or otherwise configure a spare RF chain to operate on a second operating bandwidth **660** corresponding to a bandwidth of the SSB **654** if the UE supports BWP operation without bandwidth restriction. In some examples, the second operating bandwidth **660** is separated from the first operating bandwidth **658** by a gap **662**. Once the SSB measurement occasion **656** ends, the UE can deactivate the spare RF chain to revert its operating bandwidth back to the first operating bandwidth **658**, thereby reducing power consumption. In this example, the network can cause interruptions **664**, **666** before and after the SSB measurement occasion **656** to allow the UE sufficient time for RF adjustment based at least in part on the indication that the UE supports BWP operation without bandwidth restriction. In addition, the network can assume that the UE can receive the SSB **604** and perform SSB measurements without any measurement gap based at least in part on the indication that the UE supports BWP operation without bandwidth restriction.

In the examples discussed above, the network assumed that the UE did not need a measurement gap to perform SSB measurements outside the active DL BWP based on the indication by the UE that it supports BWP operation without bandwidth restriction. In some examples, the UE can expressly indicate whether a measurement gap is needed. For instance, if a UE indicates that it supports BWP operation without bandwidth restriction (e.g., using the bwp-WithoutRestriction IE), and indicates that no measurement gap is needed using the NeedForGapsIntraFreq-r16 IE (e.g., as defined in 3GPP TS 38.331), then the network can determine that the UE supports BWP operation without bandwidth restriction and is capable of performing intra-frequency or serving cell measurement (among other RRM activities) without a measurement gap. In this case, the UE can be configured for BWP operation without bandwidth restriction as discussed, for example, with reference to FIGS. **5A-5B** and **6A-6B**. On the other hand, if a UE indicates that it supports BWP operation without bandwidth restriction, and indicates that a measurement gap is needed using the NeedForGapsIntraFreq-r16 IE, then the network can determine that the UE supports BWP operation without bandwidth restriction, but needs a measurement gap for intra-frequency or serving cell measurements (among other RRM activities). In some examples, since the UE is not required to transmit or receive data or control information during the measurement gap, the UE can adjust the RF chain during the measurement gap such that separate interruptions by the network are not needed. If a UE does not indicate that it supports BWP operation without restriction within a particular frequency layer or band, then the UE can drop or otherwise not perform measurements (e.g., intra-frequency or serving cell measurements, among other) on a SSB

outside the DL BWP and within the particular frequency layer or band, and the network can avoid configuring the UE with such a SSB (or associated serving cell) on the particular frequency layer or band.

The techniques described here can be applied to other RRM procedures, including (but not limited to) SCell activation. For example, if a UE indicates that it does not support BWP operation without bandwidth restriction, then the UE may drop a SCell activation procedure if the SSB of the target SCell is not in the first active DL BWP or any DL BWP of the target SCell. On the network side, the network may configure the UE to activate a target SCell only when the SSB of the target SCell is in the first active DL BWP of the target SCell if the UE indicates that it does not support BWP operation without bandwidth description. On the other hand, if a UE indicates that it supports BWP operation without bandwidth restriction, and the SSB of the target SCell is not in the first active DL BWP or any DL BWP of the target SCell, then the UE can be configured in accordance with the techniques described here to receive and perform measurements on the SSB. For instance, in some examples, the UE would not perform RF chain adjustment (e.g., would not adjust the operating bandwidth or turn on/off a spare RF chain) after DL synchronization for the target SCell (including automatic gain control (AGC) estimation (if needed), PSS/SSS detection (if needed), and/or time/frequency tracking) during the SCell activation procedure period (e.g., the period defined in 3GPP TS 38.133, section 8.3.2). In other words, in this example, the UE would configure an operating bandwidth that includes the first active BWP and the bandwidth of the target SSB (e.g., as discussed with reference to FIGS. 5A and 5B), and would not cause interruption to further adjust the RF chain after DL synchronization for the target SCell during the SCell activation procedure.

In some examples, the UE would perform RF chain adjustment (e.g., would adjust the operating bandwidth or turn on/off a spare RF chain) after DL synchronization for the target SCell (including AGC estimation (if needed), PSS/SSS detection (if needed), and/or time/frequency tracking) during the SCell activation procedure period, since the UE only uses an adjusted (extended) bandwidth or spare RF chain during the DL synchronization period. In other words, in this example, the UE would cause an interruption after DL synchronization for the target SCell during the SCell activation procedure period (e.g., as discussed with reference to FIGS. 6A and 6B). In this example, the network can assume that the additional interruption due to the UE RF adjustment would occur after the DL synchronization for the target SCell during the SCell activation procedure period.

In some examples, the UE would not perform RF chain adjustment (e.g., would not adjust the operating bandwidth or turn on/off spare RF chain) after DL synchronization for the target SCell (including AGC estimation (if needed), PSS/SSS detection (if needed), and/or time/frequency tracking) during the SCell activation procedure period when the UE indicates no measurement gap is needed (e.g., using the NeedForGapsIntraFreq-r16 IE) for the target intra-frequency or SCell. In other words, in this example, the UE would not cause interruption after DL synchronization for the target SCell during the SCell activation procedure. In some examples, the UE would perform RF chain adjustment after DL synchronization for the target SCell during the SCell activation procedure period when the UE indicates that a measurement gap is needed for the target intra-frequency or SCell. In other words, in this example, the UE would cause interruption after DL synchronization for the

target SCell during the SCell activation procedure period. In this example, the network can assume that the additional interruption due to the UE RF adjustment would occur after DL synchronization for the target SCell during the SCell activation procedure.

FIG. 7 illustrates an example process 700 for enhanced RRM for SSBs outside an active BWP in accordance with an aspect of the present disclosure. In some examples, the process 700 is performed by one or more of the devices or systems described here, such as the UE 101 or device 300, or the RAN node 111 or system 200, that include at least one processor (e.g., application circuitry 205, 305, baseband circuitry 210, 310, etc.) and at least one computer-readable storage medium (e.g., memory circuitry 220, 320) storing instructions executable by the at least one processor to perform the process 700.

Operations of the process 700 include communicating 702, to an access node, a capability of a UE to support BWP without restriction. For example, a UE, such as the UE 101 or device 300, can transmit (e.g., using one or more RFEMs 315) to an access node, such as the RAN 111 or system 200, an indication of its capability to support BWP without restriction. In some examples, communicating to the access node the capability of the UE to support BWP without restriction includes sending a bwp-WithoutRestriction IE to the access node.

At 704, the UE can determine that a SSB is outside a DL BWP configured for the UE. In some examples, the UE can determine that the SSB is outside the DL BWP configured for the UE based on an indication from the access node. In some examples, the access node can indicate a center frequency or other frequency (or frequency range) for the SSB.

Based at least in part on determining that the SSB is outside the DL BWP configured for the UE, the UE can adjust 706 at least one RF communication parameter to adjust its operating bandwidth. For example, the UE can adjust at least one RF communication parameter of one or more of the RFEMs 315. In some examples, adjusting the at least one RF communication parameter of the UE to adjust the operating bandwidth of the UE includes configuring the operating bandwidth to include the DL BWP and an additional bandwidth that includes the SSB. To configure the operating bandwidth to include the DL BWP and the additional bandwidth that includes the SSB, the UE can determine the additional bandwidth that includes the SSB, and expand the operating bandwidth from a first bandwidth that includes the DL BWP (e.g., an active BWP 502) to a second bandwidth that includes the DL BWP and the additional bandwidth (e.g., an adjusted operating bandwidth 506).

In some examples, adjusting the at least one RF communication parameter of the UE to adjust the operating bandwidth of the UE includes configuring a second operating bandwidth for the UE (e.g., the second operating bandwidth 558) that is in addition to a first operating bandwidth of the UE (e.g., the first operating bandwidth 556), in which the first operating bandwidth includes the DL BWP and the second operating bandwidth includes the SSB. To configure the second operating bandwidth, the UE can determine an additional bandwidth that includes the SSB, and configure the second operating bandwidth to include the additional bandwidth (e.g., using a spare RF chain or RFEM 315).

In some examples, adjusting the at least one RF communication parameter of the UE to adjust the operating bandwidth of the UE includes adjusting the at least one RF communication parameter upon occurrence of a SSB measurement window for the SSB (e.g., an SSB measurement

occasion **606**, **656**). In some examples, the operating bandwidth of the UE prior to the occurrence of the SSB measurement window can correspond to a first bandwidth (e.g., the first operating bandwidth **608**), and, upon expiration of the SSB measurement window, the UE can readjust the at least one RF communication parameter to revert the operating bandwidth of the UE to the first bandwidth.

In some examples, the UE can communicate, to the access node, an indication that a measurement gap is needed by the UE for SSB measurement. For instance, the UE can transmit to the access node a NeedForGapsIntraFreq-r16 IE to indicate that the measurement gap is needed by the UE. In this example, adjusting the at least one RF communication parameter of the UE to adjust the operating bandwidth of the UE can include adjusting the at least one RF communication parameter upon occurrence of the measurement gap.

At **708**, the UE receives the SSB using the adjusted operating bandwidth. The UE can process the SSB to perform one or more RRM activities, such as those described herein.

In some examples, the SSB received using the adjusted operating bandwidth is associated with a target SCell, and the adjusted operating bandwidth is not readjusted during a SCell activation procedure period for the target SCell. In some examples, the SSB received using the adjusted operating bandwidth is associated with a target SCell, and the operating bandwidth of the UE prior to adjustment corresponds to a first bandwidth, and the UE readjusts the at least one RF communication parameter to revert the operating bandwidth of the UE to the first bandwidth during a SCell activation procedure period for the target SCell.

FIG. **8** is a block diagram illustrating components, according to some examples, able to read instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium) and perform any one or more of the methodologies discussed herein. Specifically, FIG. **8** shows a diagrammatic representation of hardware resources **800** including one or more processors (or processor cores) **810**, one or more memory/storage devices **820**, and one or more communication resources **830**, each of which may be communicatively coupled via a bus **840**. For examples where node virtualization (e.g., NFV) is utilized, a hypervisor **802** may be executed to provide an execution environment for one or more network slices/sub-slices to utilize the hardware resources **800**.

The processors **810** may include, for example, a processor **812** and a processor **814**. The processor(s) **810** may be, for example, a central processing unit (CPU), a reduced instruction set computing (RISC) processor, a complex instruction set computing (CISC) processor, a graphics processing unit (GPU), a DSP such as a baseband processor, an ASIC, an FPGA, a radio-frequency integrated circuit (RFIC), another processor (including those discussed herein), or any suitable combination thereof.

The memory/storage devices **820** may include main memory, disk storage, or any suitable combination thereof. The memory/storage devices **820** may include, but are not limited to, any type of volatile or nonvolatile memory such as dynamic random access memory (DRAM), static random access memory (SRAM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), Flash memory, solid-state storage, etc.

The communication resources **830** may include interconnection or network interface components or other suitable devices to communicate with one or more peripheral devices

804 or one or more databases **806** via a network **808**. For example, the communication resources **830** may include wired communication components (e.g., for coupling via USB), cellular communication components, NFC components, Bluetooth® (or Bluetooth® Low Energy) components, Wi-Fi® components, and other communication components.

Instructions **850** may comprise software, a program, an application, an applet, an app, or other executable code for causing at least any of the processors **810** to perform any one or more of the methodologies discussed herein. The instructions **850** may reside, completely or partially, within at least one of the processors **810** (e.g., within the processor's cache memory), the memory/storage devices **820**, or any suitable combination thereof. Furthermore, any portion of the instructions **850** may be transferred to the hardware resources **800** from any combination of the peripheral devices **804** or the databases **806**. Accordingly, the memory of processors **810**, the memory/storage devices **820**, the peripheral devices **804**, and the databases **806** are examples of computer-readable and machine-readable media.

It is well understood that the use of personally identifiable information should follow privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining the privacy of users. In particular, personally identifiable information data should be managed and handled so as to minimize risks of unintentional or unauthorized access or use, and the nature of authorized use should be clearly indicated to users.

The methods described here may be implemented in software, hardware, or a combination thereof, in different implementations. In addition, the order of the blocks of the methods may be changed, and various elements may be added, reordered, combined, omitted, modified, and the like. Various modifications and changes may be made as would be obvious to a person skilled in the art having the benefit of this disclosure. The various implementations described here are meant to be illustrative and not limiting. Many variations, modifications, additions, and improvements are possible. Accordingly, plural instances may be provided for components described here as a single instance. Boundaries between various components, operations and data stores are somewhat arbitrary, and particular operations are illustrated in the context of specific illustrative configurations. Other allocations of functionality are envisioned and may fall within the scope of claims that follow. Finally, structures and functionality presented as discrete components in the example configurations may be implemented as a combined structure or component. These and other variations, modifications, additions, and improvements may fall within the scope of implementations as defined in the claims that follow.

What is claimed is:

1. A method, comprising:

communicating, to an access node, a capability of a user equipment (UE) to support bandwidth part (BWP) without restriction, wherein communicating to the access node the capability of the UE to support BWP without restriction comprises sending a bwp-WithoutRestriction information element (IE) to the access node;

determining that a synchronization signal block (SSB) is outside a downlink (DL) BWP configured for the UE; based at least on determining that the SSB is outside the DL BWP configured for the UE, adjusting at least one radio frequency (RF) communication parameter of the UE to adjust an operating bandwidth of the UE; and

31

receiving the SSB using the adjusted operating bandwidth.

2. The method of claim 1, wherein adjusting the at least one RF communication parameter of the UE to adjust the operating bandwidth of the UE comprises configuring the operating bandwidth to include the DL BWP and an additional bandwidth that includes the SSB.

3. The method of claim 2, wherein configuring the operating bandwidth to include the DL BWP and the additional bandwidth that includes the SSB comprises:

determining the additional bandwidth that includes the SSB; and

expanding the operating bandwidth from a first bandwidth that includes the DL BWP to a second bandwidth that includes the DL BWP and the additional bandwidth.

4. The method of claim 1, wherein adjusting the at least one RF communication parameter of the UE to adjust the operating bandwidth of the UE comprises:

configuring a second operating bandwidth for the UE that is in addition to a first operating bandwidth of the UE, wherein the first operating bandwidth includes the DL BWP and the second operating bandwidth includes the SSB.

5. The method of claim 4, wherein configuring the second operating bandwidth comprises:

determining an additional bandwidth that includes the SSB; and

configuring the second operating bandwidth to include the additional bandwidth.

6. The method of claim 1, wherein adjusting the at least one RF communication parameter of the UE to adjust the operating bandwidth of the UE comprises adjusting the at least one RF communication parameter upon occurrence of a SSB measurement window for the SSB.

7. The method of claim 6, wherein the operating bandwidth of the UE prior to the occurrence of the SSB measurement window corresponds to a first bandwidth, the method further comprising:

upon expiration of the SSB measurement window, readjusting the at least one RF communication parameter to revert the operating bandwidth of the UE to the first bandwidth.

8. The method of claim 1, comprising:

communicating, to the access node, an indication that a measurement gap is needed by the UE for SSB measurement,

wherein adjusting the at least one RF communication parameter of the UE to adjust the operating bandwidth of the UE comprises adjusting the at least one RF communication parameter upon occurrence of the measurement gap.

9. The method of claim 8, wherein communicating to the access node the indication that the measurement gap is needed by the UE comprises communicating to the access node a requirement for the measurement gap using a Need-ForGapsIntraFreq-r16 information element (IE).

10. The method of claim 1, wherein the SSB received using the adjusted operating bandwidth is associated with a target secondary cell (SCell), and wherein the adjusted operating bandwidth is not readjusted during a SCell activation procedure period for the target SCell.

11. The method of claim 1, wherein the SSB received using the adjusted operating bandwidth is associated with a target secondary cell (SCell), and wherein the operating bandwidth of the UE prior to adjustment corresponds to a first bandwidth, the method further comprising:

32

readjusting the at least one RF communication parameter to revert the operating bandwidth of the UE to the first bandwidth during a SCell activation procedure period for the target SCell.

12. The method of claim 1, comprising determining that the SSB is outside the DL BWP configured for the UE based on an indication from the access node, the indication including a center frequency for the SSB.

13. A device, comprising:

at least one processor; and

at least one computer-readable storage device storing instructions executable by the at least one processor to perform operations comprising:

communicating, to an access node, a capability of the device to support bandwidth part (BWP) without restriction;

determining that a synchronization signal block (SSB) is outside a downlink (DL) BWP configured for the device;

based at least on determining that the SSB is outside the DL BWP configured for the device, adjusting at least one radio frequency (RF) communication parameter of the device to adjust an operating bandwidth of the device;

receiving the SSB using the adjusted operating bandwidth, wherein the SSB received using the adjusted operating bandwidth is associated with a target secondary cell (SCell), and wherein the operating bandwidth of the device prior to adjustment corresponds to a first bandwidth; and

readjusting the at least one RF communication parameter to revert the operating bandwidth of the device to the first bandwidth during a SCell activation procedure period for the target SCell.

14. The device of claim 13, wherein adjusting the at least one RF communication parameter of the device to adjust the operating bandwidth of the device comprises configuring the operating bandwidth to include the DL BWP and an additional bandwidth that includes the SSB.

15. The device of claim 13, wherein adjusting the at least one RF communication parameter of the device to adjust the operating bandwidth of the device comprises:

configuring a second operating bandwidth for the device that is in addition to a first operating bandwidth of the device,

wherein the first operating bandwidth includes the DL BWP and the second operating bandwidth includes the SSB.

16. A method, comprising:

receiving, from a device, an indication of a capability of the device to support bandwidth part (BWP) without restriction;

in response to the indication, configuring the device to receive a synchronization signal block (SSB) that is outside a downlink (DL) BWP configured for the device, the configuring including causing the device to adjust at least one radio frequency (RF) communication parameter of the device to adjust an operating bandwidth of the device; and

causing an interruption period before and after a measurement occasion for the SSB that is outside the DL BWP configured for the device based at least in part on the indication of the capability of the device to support BWP without restriction.

17. The method of claim 16, wherein receiving the indication of the capability of the device to support BWP

without restriction comprises receiving a bwp-With-
outRestriction information element (IE) from the device.

18. The method of claim **16**, comprising:

receiving, from the device, an indication that a measure-
ment gap is needed by the device for receipt of the SSB 5
that is outside the DL BWP configured for the device;
and

scheduling the measurement gap for the device before and
after a measurement occasion for the SSB that is
outside the DL BWP configured for the device. 10

* * * * *